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Chapter 12

Plant Growth and Development

Chapter Contents

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Development is the sum of two processes: growth and differentiation. To begin with, it is essential and sufficient to know that a mature plant develops from a single-celled zygote which divides and differentiates to form various types of cell. These different types of cells organise to form tissues and organs of the plant. The development of a mature plant from a zygote follows a precise and highly ordered succession of events. During this process, a complex body *i.e.*, plant is formed which constitutes the roots, leaves, branches, flowers, fruits and seeds.

GROWTH

Growth is a fundamental characteristic of all living organisms. It is defined as an irreversible permanent increase in size, volume and weight of an organ or its parts or even of an individual cell.

Plant Growth is Generally Indeterminate

Plants have the capacity for indeterminate or unlimited growth which continues throughout their life. Due to which, **the growth of a plant is known as unique**. The growth of the plants is due to the presence of different types of meristems at specific locations in their body. Meristems are the cells which are capable of self-perpetuation *i.e.*, have the power to continue indefinitely through division. The continued growth due to which new cells are always being added to the plant body by the activity of the meristem is called the **open form of growth**.

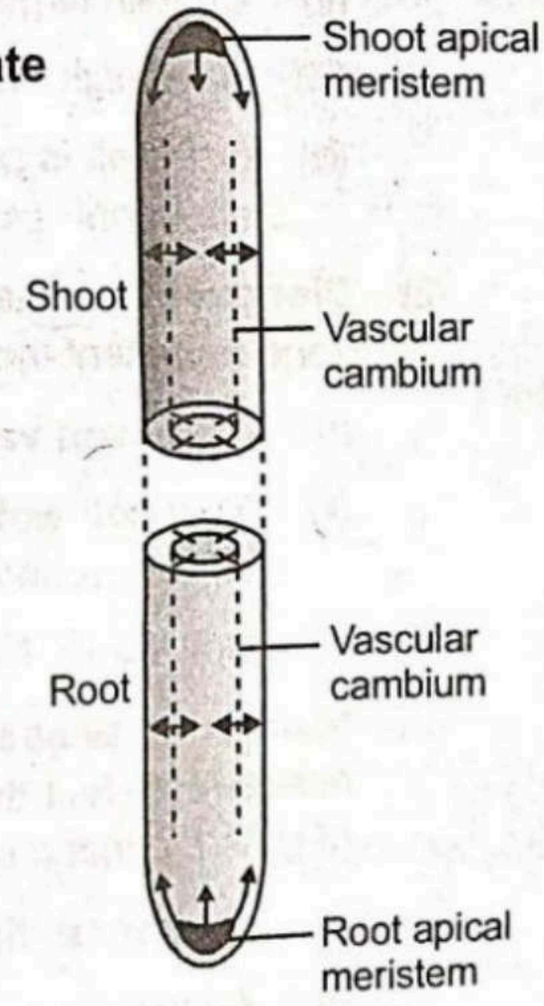


Fig.: Diagrammatic representation of locations of root apical meristem, shoot apical meristem and vascular cambium. Arrows exhibit the direction of growth of cells and organ.

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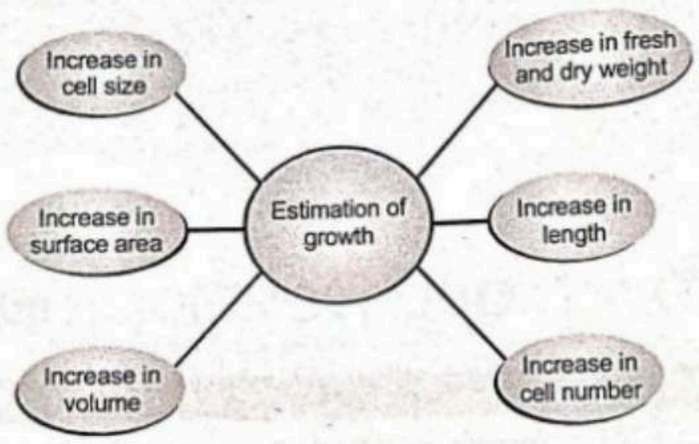
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Growth is Measurable

Growth can be measured by various parameters, as shown below:



Auxanometer is an instrument used to measure the growth in length of the plant.

Phases of Growth

Plant growth takes place in three phases :

(1) **Meristematic phase or phase of cell division :** It is the first phase of growth in plants which occurs in the areas where meristematic cells are present. For example, at shoot and root tip of the plants where meristems are constantly dividing. As a result this phase is also known as cell division.

The cells at this phase of growth show following features :

(i) Have dense protoplasm.

(ii) Contain large nucleus.

(iii) Have high respiration rate.

(iv) Cell wall is primary in nature. It means that the cell wall is made up of cellulose, thin walls with abundant plasmodesmatal connections so that cells can communicate with each other.

(2) **Elongation phase or phase of enlargement :** The cells present just next to the cells of meristematic zone represent the elongation phase. The cells found in this zone show the following features :

(i) **Increased vacuolation :** The number and size of vacuoles increase as growth occurs.

(ii) **New cell wall deposition :** The cell wall of the cells starts depositing new materials resulting in cell enlargement.

(iii) **Cell size :** The size of the cells of this phase increases mainly due to vacuolation.

(3) **Maturation phase or phase of differentiation :** The cells present just next to the cells of elongation phase represent the phase of differentiation. The enlarged cells develop into special or particular type of cells by undergoing structural and physiological differentiation.

(i) **Structural differentiation :** A cell attains a particular shape, size and internal constitution.

(ii) **Physiological differentiation :** A cell attains a particular function. For example, absorption of water and minerals by root hair, photosynthesis by mesophyll cells etc.

So, cells of this zone, attain their maximal size in terms of wall thickening and protoplasmic modifications.

After differentiation, the mature cells do not grow further and remain unchanged till death.

Fig. : Detection of zones of elongation by the parallel line technique. Zones A, B, C, D immediately behind the apex have elongated most.

Growth Rates

An organism or a part of the organism can produce more cells in a variety of ways. The increase in growth per unit time is defined as growth rate. Thus, it can be expressed mathematically, being either arithmetical or geometrical.

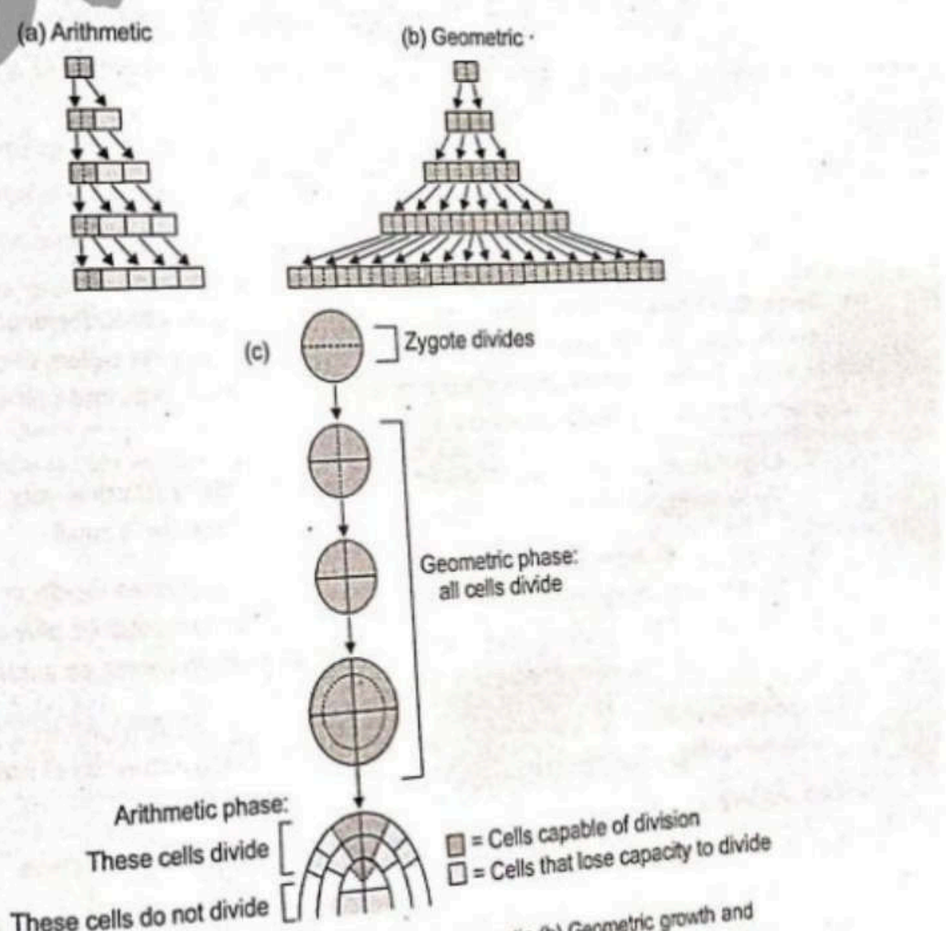


Fig. : Diagrammatic representation of : (a) Arithmetic (b) Geometric growth and (c) Stages during embryo development showing geometric and arithmetic phases.

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(i) **Arithmetic growth :** It is the growth rate in which growth occurs at a constant rate from the very beginning and progresses arithmetically. Here, following mitotic cell division only one daughter cell divides continuously whereas other cells undergo differentiation and become mature. For example, we can study the arithmetic growth pattern in root elongation where elongation occurs at a constant rate. If a graph is plotted by taking length of the organ of plant at Y-axis against the time at X-axis, a linear curve is obtained. Mathematically, it can be expressed by following equation :

$$L_t = L_0 + rt$$

where L_t = Length of the organ at time 't'

L_0 = Length of the organ at time 'zero'

r = Growth rate or elongation per unit time.

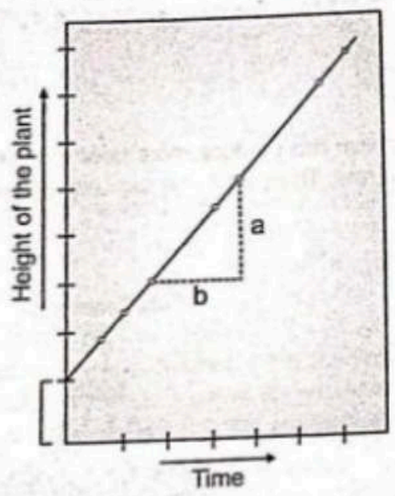


Fig. : Constant linear growth, a plot of length L against time t

(ii) **Geometrical growth :** It is a growth rate where every cell divides with all the daughters growing and dividing again. We can take the example of microorganisms to study the pattern of geometrical growth. In microorganisms, growth occurs when they are provided with enough food and space. During their growth, three phases can be observed .

(1) **Lag phase :** This is the initial phase of growth when the rate of growth is very slow. It represents the beginning of growth of microorganisms where their cell number is small.

(2) **Log phase or exponential phase :** In this phase, growth progresses rapidly or exponentially *i.e.*, reaches to its maximum. Here, both the progeny or daughter cells obtained after mitotic cell division have the ability to divide continuously. However, such a growth cannot be sustained for long and growth reaches to next phase.

(3) **Stationary phase :** Due to the shortage of space, food and accumulation of toxins, growth slows down leading to a phase known as stationary phase.

If we plot all these phases graphically then we will obtain S-shaped or sigmoid curve.

Similarly, we can study the growth of cells, tissues and organs of a plant. For example, if we plot the parameters of growth such as size, weight of the organ at Y-axis against the time at X-axis, we get a typical sigmoid or S-shaped curve which explains the geometrical growth.

But a tree showing seasonal activities does not show a typical S-shaped curve. Here, the curve will show small steps indicating stoppage and resumption of growth every year.

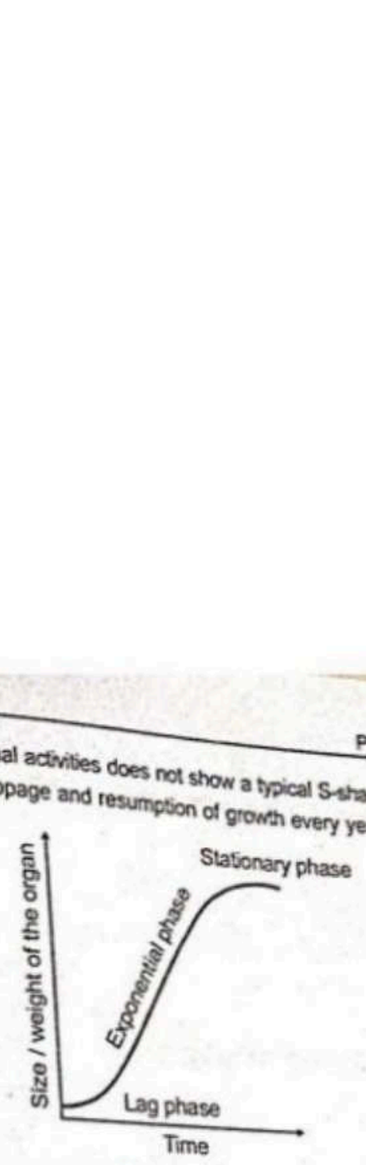


Fig. : An idealised sigmoid growth curve typical of cells in culture, and many higher plants and plant organs

$$W_t = W_0 e^{rt}$$

where W_t = Final size (weight, height, number etc.); W_0 = Initial size at the beginning of the growth; e = Base of natural logarithms; r = Growth rate; t = Time of growth

The 'r' expressed in the above equation is actually the relative growth rate which measures the ability of the plant to produce or synthesize new plant material. This ability of plant is referred as **efficiency index**. The overall size of any organ of the plant depends on its initial size *i.e.*, W_0.

There are two ways by which quantitative comparisons between the growth of living system can be made :

(i) Absolute growth rate

(ii) Relative growth rate

Absolute growth rate : It is the measurement and comparison of total growth per unit time.

Relative growth rate : The growth of the given system per unit time expressed on a common basis, example, per unit initial parameter is called the relative growth rate.

$$\text{Relative growth rate} = \frac{\text{Growth per unit time}}{\text{Initial size}} \times 100$$

For example, two leaves, A and B of different size show an absolute increase in area at the given time to give leaves A^1 and B^1.

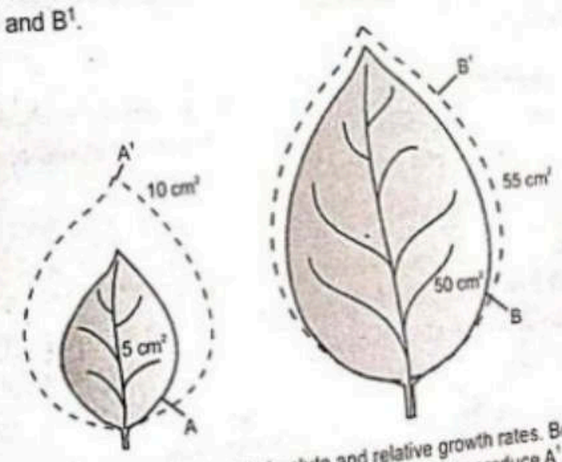


Fig. : Diagrammatic comparison of absolute and relative growth rates. Both leaves A and B have increased their area by 5 cm^2 in a given time to produce A', B' leaves.

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The absolute growth rate of both the leaves A^1 and B^1 is same but if we calculate the relative growth rate, it is found that A^1 leaf has higher relative growth rate than B^1 leaf which can be proved by the following calculations.

Relative growth rate (RGR) of A^1 leaf $= \frac{5}{5} \times 100$

$= 100\%$ ∵ Growth per unit time = 5 cm^2

Initial size = 5 cm^2

Relative growth rate (RGR) of B^1 leaf $= \frac{5}{50} \times 100$

$= 10\%$ ∵ Growth per unit time = 5 cm^2

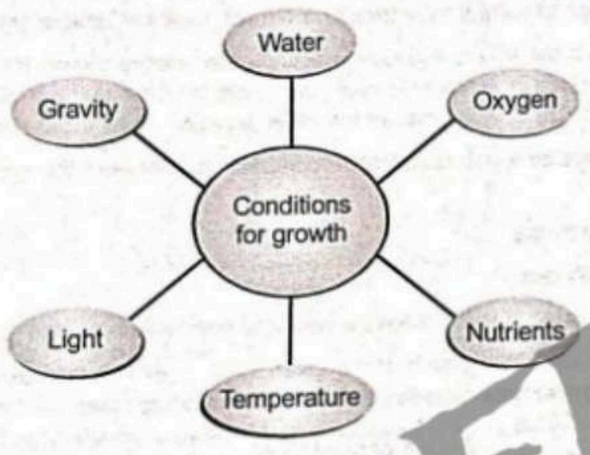
Initial size = 50 cm^2

A^1 leaf has 100% relative growth rate whereas B^1 leaf has 10% relative growth rate.

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Conditions for Growth

Growth is influenced by various factors such as:



DIFFERENTIATION, DEDIFFERENTIATION AND REDIFFERENTIATION

Differentiation

The phenomenon by which the cells derived from root apical meristem, shoot apical meristem undergo permanent changes in their structure, biochemistry, size, physiology of cell wall and protoplasm content. Thus, enabling the cells to perform a specific function.

Some examples of the differentiation are :

(i) **Tracheary element :** The tracheary element is formed by the process of differentiation where the cells elongate and loose their protoplasm to form tracheids. A strong, elastic lignocellulosic secondary cell wall develops which does not allow tracheids to collapse under extreme tension and allow it to carry water to long distances.

(ii) **Chlorenchyma :** It is specialised to perform photosynthesis which is developed due to formation of chloroplasts in the living thin-walled cells.

Other structures found in plants such as aerenchyma, stomata, collenchyma, trichomes, root cap etc. are formed after the differentiation.

Dedifferentiation

During differentiation cells loose their ability to divide and form permanent cells. But some cells regain their capacity to divide under certain conditions. This phenomenon where certain living differentiated cells regain or attain their ability to divide and form new cells is known as dedifferentiation.

A dedifferentiated tissue can act as a meristem *e.g.*, interfascicular vascular cambium, cork cambium, wound cambium. These are formed from fully differentiated parenchyma cells.

Redifferentiation

The dedifferentiated cells can again loose their ability to divide to form permanent cells specialised for a particular function. The process where the dedifferentiated cells again loose their ability to divide to form permanent cells is called redifferentiation. It is similar to differentiation of cells and tissues formed by primary meristem. (For examples – Secondary phloem, secondary xylem, cork, secondary cortex are some tissues formed through redifferentiation.)

Like growth, differentiation is also open in plants. The same apical meristem produces parenchyma, fibres, xylem, phloem, collenchyma and epidermis. The final structure of a cell, formed at maturity is also determined by the neighbouring cells present in the tissue. For example, cells positioned away from root apical meristem differentiate as root cap cells, while those pushed to the periphery mature as epidermis.

DEVELOPMENT

Development is the sequence of changes that occurs in the structure and functioning of an organism, organ, tissue or cell involving its formation, growth, differentiation, maturation, reproduction, senescence and death. Plant during its life cycle passes through development stages of seed germination, seedling, juvenile phase, maturation, flowering, seed formation and senescence. The conversion of one phase into next is also development *e.g.*, leaf initiation to leaf expansion, vegetative or juvenile phase to reproductive phase *i.e.*, flowering. Appearance of chloroplasts in cells when exposed to sunlight is also a type of development. Development leads ultimately to senescence and then death.

Developmental Stages of Plant

Seed germination
↓
Seedling
↓
Juvenile phase
↓
Maturation
↓
Flowering
↓
Seed formation
↓
Senescence

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Fig. : Sequence of the developmental processes in a plant cell.

The development process is not always straight. Plants show **plasticity** which means the ability of plant to follow different pathways and produce different structures in response to environment. Example of plasticity is **heterophylly**. Heterophylly is the phenomenon of appearance of different forms of leaves on the same plant. It can be seen or observed in plants like cotton, coriander and larkspur. In such plants, the leaves of the juvenile plants are different in shape from those of mature plants. On the other hand, difference in shapes of leaves can be observed in buttercup present in air and water. It means this plant shows different shapes of leaves according to its habitat.

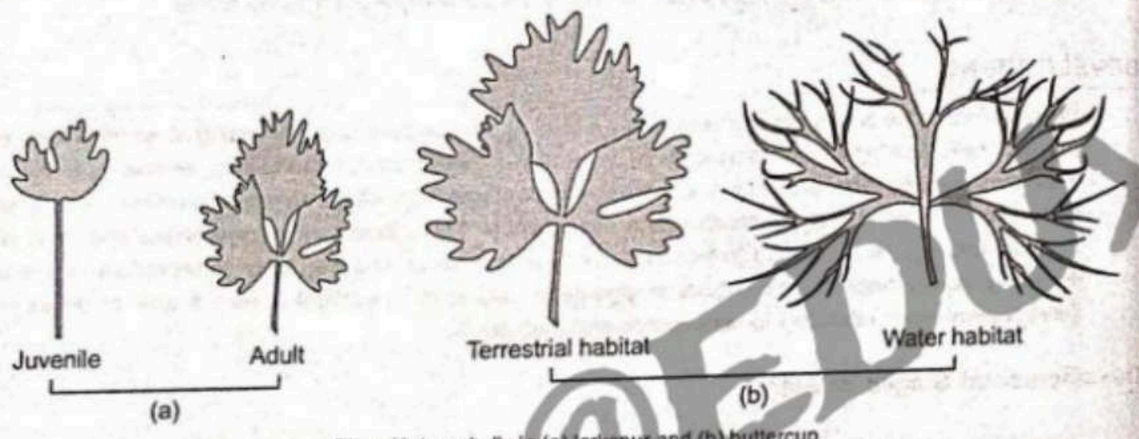
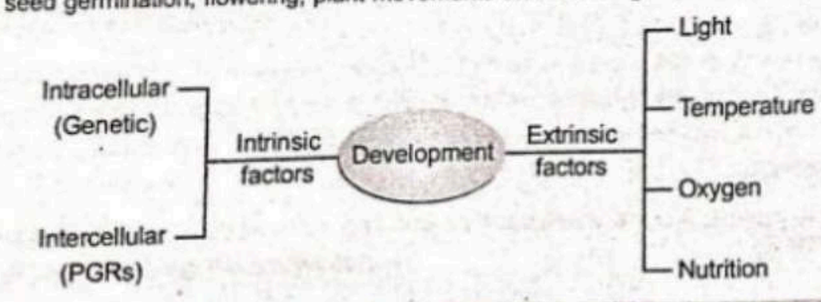


Fig. : Heterophylly in (a) larkspur and (b) buttercup

It is clear, therefore, that development includes both growth and differentiation. There are various intrinsic and extrinsic factors which influence the developmental stages of a plant. The intrinsic factors include genetic which are intracellular and chemical substances like PGRs which are intercellular. The extrinsic or external factors are light, temperature, oxygen, CO_2, water, nutrition etc. These factors influence various events such as dormancy, seed germination, flowering, plant movements etc. occurring in plants during their development.



PLANT GROWTH REGULATORS

Plant growth regulators (PGRs) are small, simple organic substances of diverse or different chemical composition which are required in low concentration like hormones in animals. The PGRs influence physiological activities of plants leading to promotion, inhibition and modification of growth. Therefore, they are also called as **plant growth substances, plant hormones or phytohormones.**

A plant hormone can be best defined as a chemical substance other than nutrients which regulate one or more physiological activities at the same site where it gets synthesised or away from that. These phytohormones are composed of different types of chemical compounds which are as follows :

S. No.	Composition	Phytohormones
1.	Indole compounds	IAA – Indole-3-acetic acid, IBA – Indole butyric acid
2.	Terpenes	Gibberellic acid or Gibberellins (GAs)
3.	Adenine derivatives	N^6 – Furfurylaminopurine, kinetin
4.	Gases	Ethylene ($CH_2=CH_2$)
5.	Derivatives of carotenoids	Abscisic acid (ABA)

Classification of PGRs / Phytohormones

On the basis of their functions performed in a living plant body, these can be broadly classified into two groups *i.e.*,

(i) **Plant growth promoters :** These are the chemical substances involved in growth-promoting activities such as cell division, cell enlargement, tropic growth, flowering, fruiting and seed formation. *e.g.*, **Auxin, gibberellins and cytokinin.**

(ii) **Plant growth inhibitors :** These are the chemical substances such as **ABA**, involved in growth-inhibiting activities like dormancy and abscission.

(a) **Dormancy :** It is a period when growth and development are temporarily stopped. For example, failure of seeds to germinate due to the lack of favourable conditions in the environment.

(b) **Abscission :** It is the falling or shedding of leaves, fruits and flowers.

One other type of phytohormone *i.e.*, **ethylene** could fit either of the two above mentioned groups, but it is largely an inhibitor of growth activities and is a gaseous hormone.

Now we will study different types of phytohormones in detail.

1. **Auxin :** The word auxin is derived from Greek word '*auxein*' which means 'to grow'.

Discovery : Auxin was the first hormone to be discovered. **Charles Darwin** and his son **Francis Darwin** observed that the stimulus of light was perceived by coleoptile tip of canary grass but the bending response to unilateral light was produced at a distance in growth zone *i.e.*, subapical part.

After a series of experiments conducted by other scientists, it was concluded that the sensation picked up by the coleoptile tip is transmitted to the subapical part which bends towards the direction of light. As it was found that :

(a) In an experiment if the coleoptile tip was removed then there was no bending of shoot towards light because the stimulus of light was not perceived.

(b) When coleoptile tip was covered by an opaque tin foil cap, it could not perceive the stimulus of light and hence, no bending was observed.

(c) When coleoptile tip was covered with a translucent cap, it could perceive the stimulus of light and hence, bending was observed.

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Fig. : Experiment used to demonstrate that tip of the coleoptile is the source of auxin. Arrows indicate direction of light.

Isolation :

(a) Auxin was first isolated from human urine samples because auxins do not have metabolic role so excreted out of the body.

(b) It was also isolated from the tips of coleoptiles of oat seedlings by **F.W. Went**.

(c) **Precursor** : Tryptophan (amino acid).

Occurrence :

They occur in the growing apices of the stems and roots from where they migrates to their site of action.

Types : Auxins are of two types :

(a) Natural auxins : These are produced by plants and can be isolated from them.

For example, IAA – Indole-3-acetic acid

IBA – Indole-3-butyric acid.

(b) Synthetic auxins : These are produced artificially.

For example, 2, 4-D – 2, 4-Dichlorophenoxyacetic acid

2, 4, 5-T – 2, 4, 5-Trichlorophenoxy acetic acid.

NAA – Naphthalene acetic acid.

All these auxins have been used extensively in agricultural and horticultural practices.

Bioassay : The functions of auxins can be tested with the help of bioassays. **Bioassay** means the testing of substance for its activity in causing a growth response in a living plant or its part. The test measures the concentration required to produce the effect, and thus, it is quantitative. The ***Avena* curvature test**, and **root growth inhibition test** are some of the bioassays for examining auxin activity.

Boysen-Jensen observed that coleoptiles with decapitated tips neither grow, nor bend towards light, (i.e., do not show phototropism). The normal bending also occured, if the tip is cut off and replaced with an intervening block of gelatin. It was thus evident that the influence was due to chemical, diffusing through the block of gelatin.

A. Paal demonstrated that Coleoptile would bend even in the dark after certain treatments.

F.W. Went cut off tips of oat coleoptiles and placed them on small agar blocks and let them for a few hours to allow all the 'chemical influence' to diffuse into the agar. By this, he isolated auxins in agar block. He then placed these agar blocks on one side of freshly decapitated coleoptiles. He observed that the coleoptiles grew and bent away from the side on which the block was placed. The degree of curvature of the coleoptile was directly proportional to the concentration of the chemical influence in the agar block. Went named this 'chemical influence' responsible for the phototropic response as **auxin**. Hence, Went is credited with the discovery of auxins.

Physiological effects / Functions of auxin :

(i) Abscission : In the presence of normal level of auxin, plant organs such as fruits, leaves cannot abscise. Thus, auxin inhibits the abscission, it can occur only when auxin content or level decreases and a layer called abscission layer is formed between the organs like fruits, flowers etc. and the stem. Thus, auxin inhibits abscission of young leaves, fruits and flowers whereas it promotes the abscission of older mature leaves, fruits and flowers.

(ii) Apical dominance : It is the phenomenon by which the presence of apical bud does not allow the nearby lateral or axillary buds to grow. Apical bud secretes auxin which inhibits the growth of lateral buds. If the apical bud is removed (**Decapitation**), the lateral buds sprout i.e., start growing. However, if a paste containing auxin is applied on the cut portion of the decapitated shoot, the lateral buds remain inhibited in the similar way when apical bud is present.

The removal of apical bud is done for the preparation of hedges and bushy growth in certain plants. For example, when the apical bud of tea plants is removed, the lateral buds start developing into branches. It is done to produce more number of leaves so that yield can be increased.

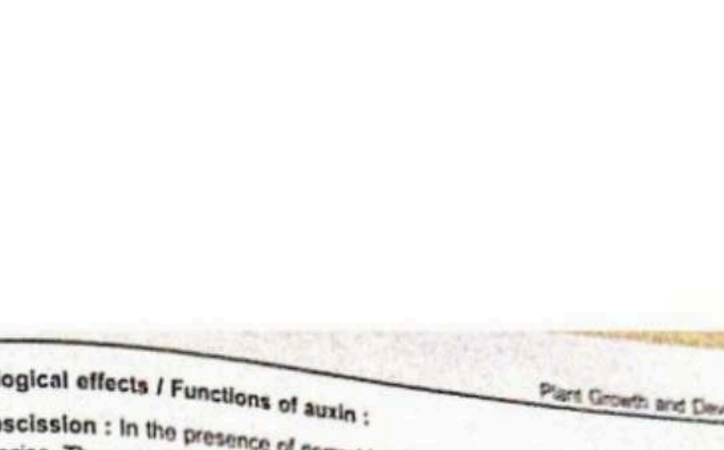


Fig. : Apical dominance and auxin. A, apical bud intact. B, removal of apical bud and loss of dominance. C, auxin applied over cut end inhibits lateral bud growths as in apical dominance.

(iii) Cell division and Xylem differentiation : Auxin is responsible for initiation and promotion of cell division in cambium. It also controls the xylem differentiation.

Uses / Applications

(1) Rooting : Auxin stimulates root formation on stem cuttings, particularly the woody ones. The common auxins used for inducing the rooting are NAA and IBA.

(2) Flowering : The dilute solutions of NAA and 2,4-D are sprayed on litchi and pineapple which induce flowering in them. However, flowering in most plants can be inhibited by spraying high concentration of auxins. This effect is used advantageously to prevent flowering in lettuce, where only leaves are edible.

(3) Parthenocarpy : Auxins such as IAA and IBA in diluted form are used to produce parthenocarpic or seedless fruits *e.g.*, tomatoes. It has been observed that the carpels (female part) producing parthenocarpic or seedless fruits like banana, grapes etc. have a higher internal production of auxins.

(4) Weedicides : Some synthetic auxins are used as weedicides. Weedicides are the chemicals used to kill weeds (unwanted plants) growing with the crops. 2, 4-D is widely used to remove broad leaved weeds or dicotyledonous weeds in cereal crops or monocotyledonous plants.

The auxins are also used by gardeners to prepare weed-free lawns. Thus, auxins are widely used in agricultural and horticultural practices.

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Note :

(i) *Synthetic auxins have also been misused by people. Large scale aerial application of weedicides used in Vietnam to expose the forests have exterminated the wild relative of Citrus and these auxins were called 'Agent orange' during Vietnam-US war.*

(ii) *Lodging or falling of many crop plants during windy seasons can be prevented by the application of NAA.*

EXERCISE

1. Which of the following statement for Plant growth is incorrect?
   (1) It is localised (2) It is open ended
   (3) It is indeterminate (4) It does not involve increase in number of parts
2. Which of the given plant shows plasticity in response to its prevailing environment?
   (1) Larkspur (2) Buttercup
   (3) Cotton (4) Coriander
3. 2, 4-D is
   (1) Widely used to kill monocots
   (2) Effective weedicides against broad leaved plants
   (3) Used to prevent lodging and root initiation
   (4) Used to promote flowering in pineapple and mango
4. Select an incorrect statement
   (1) Auxin promotes xylem differentiation
   (2) NAA is used to overcome apical dominance
   (3) Auxins help to prevent fruit and leaf drop at early stages
   (4) Auxins prevent lodging of cereals

5. Auxins
   (1) Were first isolated in agar by Paal (2) Are synthesised using tryptophan as precursor
   (3) Are found always as bound with amines (4) Are synthesised in shoot apices only

(2) Gibberellins

Discovery : The effect of gibberellins had been known in Japan for over a century ago. Japanese farmers observed that certain rice seedlings in their fields grow excessively tall and become weak and sterile. It was found that it is a type of disease from which the seedlings suffered known as 'bakane' or 'foolish seedling' disease.

Japanese plant pathologist **E. Kurosawa** (1926) found that the disease was caused by a fungus, *Gibberella fujikuroi*. With the extract of this fungus, he could produce symptoms of foolish seedling in healthy plants.

Yabuta extracted the active substances *i.e.*, growth-promoting substances and termed them as gibberellin.

Presently, more than 100 gibberellins have been identified from widely different organisms such as fungi and higher plants. They are acidic in nature and denoted as GA_1, GA_2, GA_3 and so on. Among all gibberellins or gibberellic acid, GA_3 is one of the first to be discovered and thoroughly the most intensively studied form.

Precursor : Acetyl-CoA

Physiological effects / Functions :

(i) **Stem and leaf growth :** Gibberellins stimulate stem elongation and other aerial parts. Therefore, these increase the size of stem and fruits. However, gibberellins have no effect on roots.

(ii) **Bolting :** Gibberellins induce stem elongation in 'rossette plants' *e.g.*, cabbage, beet. In rossette plants, internodal growth is retarded and leaves are developed profusely. In these plants, just prior to the reproductive phase, the internodes elongate enormously causing a marked increase in stem height. This is known as bolting.

(iii) **Seed germination :** Gibberellins stimulate the synthesis of various types of hydrolytic enzymes for mobilisation of reserve food, *e.g.*, amylases, proteases.

Uses / Application:

(1) **Fruit growth :** Gibberellins are used to increase the size of apple as well as improve its shape and increases bunch length in grapes.

(2) **Delayed ripening :** With the help of gibberellins, ripening of fruits can be delayed so that their plucking can be postponed. Thus, fruits can be left on the tree longer so as to extend the period for marketing.

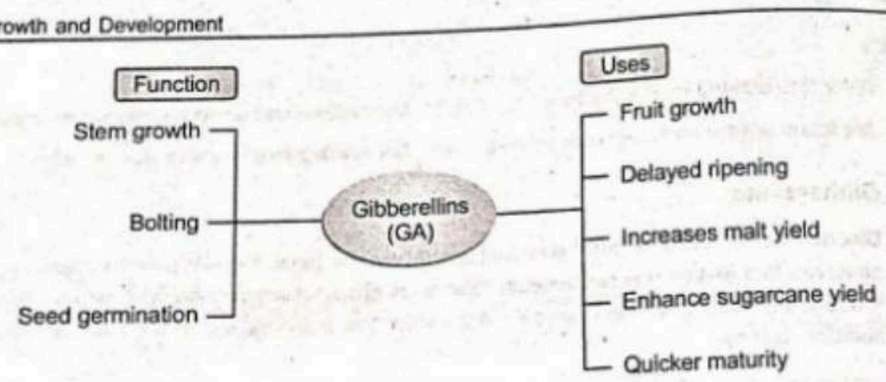
(3) **Malt yield :** GA_3 is used to speed up the malting process in brewing industry. It increases the yield of malt from barley grains.

(4) **Sugarcane :** Carbohydrate in the form of sugar is stored in the stems of sugarcane. When gibberellin is sprayed on sugarcane crop, the length of the stem increases. As a result, the sugar content increases which finally increases the yield by as much as twenty tonnes per acre.

(5) **Quicker maturity :** In juvenile conifers, gibberellin has been found to cause quicker early growth so that maturity is reached early. It is useful for obtaining early and quicker yield of economically important seeds.

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Note :

A. Bioassay of Gibberellins

(i) *Induction of α-amylase in barley endosperm test.*

(ii) *Dwarf maize test.*

(iii) *Dwarf pea test.*

B. Anti-GA:

(i) *CCC (Chlorocholine chloride)*

(ii) *AMO1618*

(iii) *Phosphon-D*

(iv) *Maleic hydrazide (MH)*

(3) Cytokinins

Discovery : F. Skoog and his co-workers found that tobacco callus *i.e.*, a mass of undifferentiated cells proliferated only when in addition with auxin, the culture medium is provided either with coconut milk or yeast extract DNA.

Miller *et al.* discovered the first cytokinin from degraded product of autoclaved herring sperm DNA which had a powerful cytokinesis (division of cytoplasm) promoting effect. It is called kinetin (N^6-furfurylamino purine). It is a synthetic product and does not occur naturally in plants.

Isolation : The first natural cytokinin was isolated from unripe maize grains or kernels known as zeatin. It can also be isolated from coconut milk. Since the discovery of zeatin, several naturally occurring cytokinins, and some synthetic compounds with cell division-promoting activity have been identified.

Occurrence : Cytokinins occur in regions where rapid cell division occurs such as root apices, developing shoot buds, young fruits etc.

Chemically these are modified purines, **derived from tRNA**. *e.g.*, zeatin (Benzyl amino purine; BAP), dihydrozeation, Isopentenyl adenine (IPA).

Functions / Physiological effects

(i) **Growth of leaves and shoot :** It helps in the production of new leaves and chloroplasts in leaves. It also stimulates lateral shoot growth and adventitious shoot formation.

(ii) **Counteracting apical dominance :** Cytokinin initiates the growth of lateral buds despite the presence of apical bud. In the presence of cytokinins, supply of water and minerals increases to lateral buds. Therefore, the lateral buds grow even in the presence of apical bud.

(iii) **Delay in senescence :** Cytokinins delay the senescence (ageing) of leaves and other organs by controlling protein synthesis and mobilisation of nutrients as resources **(Richmond Lang effect)**.

(iv) **Cell division :** Cytokinins are essential for cytokinesis *i.e.*, division of cytoplasm. Along with auxin, cytokinin causes division even in permanent cells. These two hormones are also responsible for cell division in callus *i.e.*, a mass of undifferentiated cells.

(v) Shelf life of cut shoot, vegetables and flowers is increased by applying this hormone.

Uses / Applications

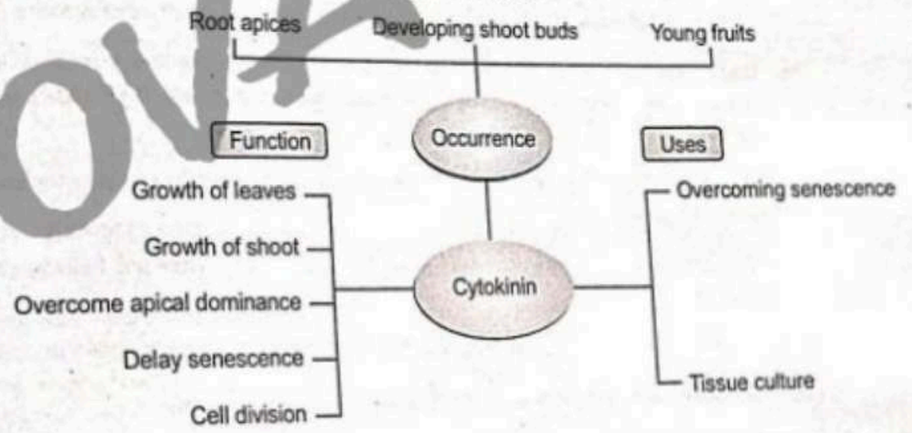
(1) **Overcoming senescence :** Cytokinins are used to delay the senescence of intact leaves and other plant parts.

(2) **Tissue culture :** Cytokinins along with auxins are essential in tissue culture as they are required for cell division and morphogenesis/organogenesis.

Auxin concentration = Cytokinin concentration → Callus

Auxin concentration > Cytokinin concentration → Root

Cytokinin concentration > Auxin concentration → Shoot



Note :

Bioassay of cytokinin :

(i) *Chlorophyll preservation test.*

(ii) *Cell division test*

(iii) *Tobacco pith culture test.*

(iv) *Expansion of excised radish cotyledon test.*

(4) Ethylene

It is a gaseous plant hormone.

Discovery : Cousins discovered and confirmed that ripe oranges emitted a volatile substance that brought about early ripening of unripe bananas kept nearby. Later on, this volatile substance was identified as ethylene.

Synthesis : It is synthesised in almost all plant parts like roots, leaves, flowers, fruits, seeds etc. The maximum synthesis occurs during ripening of fruits and in the tissues undergoing senescence *i.e.*, ageing.

Precursor : Methionine

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Physiological effects / Functions

(i) **Transverse or horizontal growth :** Ethylene inhibits longitudinal growth but stimulates transverse growth of seedlings. It is also responsible for the formation of apical hook in dicot seedling and swelling of the axis due to which stem looks swollen.

(ii) **Senescence :** It promotes senescence of plant parts like leaves, flowers. It has been found that leaf senescence is largely caused by ethylene.

(iii) **Abscission :** Ethylene accelerates abscission *i.e.*, shedding of leaves, flowers and fruits.

(iv) **Fruit ripening :** Ethylene is a ripening agent. It is involved in ripening of **climacteric fruits**. The climacteric fruits are fleshy fruits which show sudden sharp rise in the rate of respiration at the time of ripening. This sudden sharp rise in the rate of respiration is called **respiratory climactic**.

(v) **Breaking of dormancy :** It breaks the dormancy of seeds, buds and initiates the germination of seeds such as peanut seeds. It also causes the sprouting of potato tubers.

(vi) **Elongation of stem in deep water rice plants :** Both stem and petiole elongation in submerged and partially submerged aquatic plants is promoted by ethylene. For example, leaves of rice seedlings remains out of water due to ethylene where it induces rapid growth of internodes and leaf bases.

(vii) **Roots and root hairs :** Ethylene induces development of adventitious roots on various types of cuttings. It promotes the development of lateral roots and growth of root hairs so as to increase the absorption surface.

(viii) **Flowering :** Flowering and fruit set up are synchronised by ethylene in pineapples. It also initiates flowering in mango.

(ix) **Sex expression :** Ethylene increases the number of female flowers and fruits in certain plants like cucumber.

(x) **Epinasty :** Exposure to ethylene causes drooping of leaves and flowers.

(xi) **Apogeotropism :** Ethylene decreases sensitivity of roots to gravity.

Uses / Applications

Ethylene is one of the most widely used PGRs in agriculture as it regulates so many physiological processes which are discussed earlier. The most widely used compound as a source of ethylene is **ethephon**. The uses of ethylene are as follows :

(1) **Fruit ripening :** Ethephon is used in artificial ripening. It is an aqueous solution which is readily absorbed and transported within the plant and releases ethylene slowly. Ethephon has the capacity to increase or hasten fruit ripening.

Fruits such as tomatoes, apple, banana, mango etc. are picked up and transported to godowns before ripening, where ethylene is used to induce artificial ripening in them.

(2) **Abscission and thinning :** It is used to accelerate or increase the abscission of flowers and fruits and also the thinning of fruits. *e.g.*, cotton, cherry, walnut.

(3) **Increase in number of fruits :** Application of ethylene in cucumber increases the number of female flowers thereby increasing the yield.

(4) **Sprouting of storage organs :** The sprouting of storage organs such as rhizomes, tubers can be enhanced by exposing them to ethylene.



Note :

Bioassay of ethylene : *Triple response test.*

(5) ABA - Abscisic acid

Abscisic acid is a mildly acidic growth hormone which functions as a general growth inhibitor.

Discovery : During mid 1960s, three independent researchers reported the purification and chemical characterisation of three different kinds of inhibitors – inhibitor β, abscission II and dormin. Later all the three were proved to be chemically identical. In order to avoid confusion the name abscisic acid was given.

Occurrence : Dormant buds, seeds.

Precursor : Violaxanthin

Physiological effects / Functions

(i) **Abscission :** ABA promotes abscission of flowers and fruits.

(ii) **Dormancy of buds and seeds :** ABA acts as a growth inhibitor and induces dormancy of buds and seeds. As ABA induces dormancy, it is also known as dormin. The buds as well as seeds sprout only when ABA is overcome by gibberellins.

(iii) **Metabolism :** ABA inhibits the protein and RNA synthesis and cause destruction of chlorophyll. As a result senescence of leaves is stimulated.

(iv) **Inhibition of seed germination :** Abscisic acid inhibits gibberellin mediated amylase formation during germination of cereal grains and thus inhibits their sprouting.

(v) **Closure of stomata :** The concentration of ABA increases in the leaves of the plants during dessication and other stressful conditions. As a result, stomata present in the epidermis of leaves close to prevent the loss of water *i.e.*, transpiration. Therefore, it can also be known as anti transpirant. Thus, synthesis of ABA is stimulated by drought, water logging and other adverse environmental conditions. Therefore, it is known as **stress hormone**.

In most situation, ABA acts as an antagonist to GAs that's why it is known as **Anti GA**.

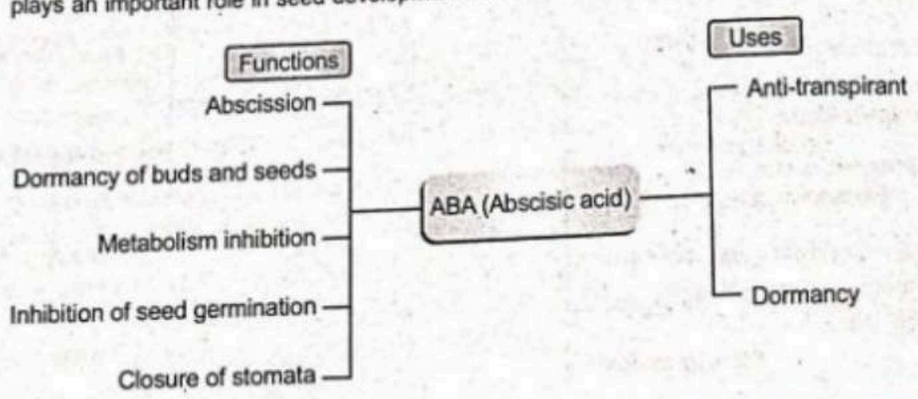
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Uses /Applications

(1) **Antitranspirant :** Application of minute quantity of ABA to leaves reduces transpiration to a great extent through partial closure of stomata. It thus, conserves water and reduces the requirement of irrigation.

(2) **Dormancy :** ABA is used to induce the dormancy of buds, seeds and storage organs. By inducing dormancy ABA helps the seed to withstand desiccation and other unfavourable factors. Thus, ABA plays an important role in seed development and maturation.



Note :

Bioassay of ABA : *Cotton or Bean explant test.*

EXERCISE

6. Which of the following plant hormone is used to speed up the malting process in brewing industry?
   (1) GA_3 (2) Ethylene
   (3) Cytokinin (4) Auxin
7. Internodal elongation in sugarcane stem is promoted by
   (1) Gibberellin (2) Cytokinin
   (3) Auxin (4) ABA
8. Select an incorrect match
   (1) ABA – Violaxanthin (2) Ethylene – Methionine
   (3) GA_3 – Acetyl CoA (4) Cytokinin – Terpenes
9. Fruit ripening is promoted by the hormone that
   (1) Promotes thinning of fruits in cherry and walnut
   (2) Induce parthenocarpy in tomato
   (3) Promotes flowering in *Lemna*
   (4) Is derived from tRNA
10. Abscisic acid is useful in
   (1) Cell division (2) Seed development
   (3) Germination of peanut seeds (4) Increasing length of stalks in grapes

In plants, growth, differentiation and developmental processes are regulated by one or other hormone. The phytohormones can act individually, synergistically or antagonistically. Synergistic action means two or more hormones have combined effect while antagonistic means two hormones work in opposite manner.

Synergistic actions

(1) **Cell division :** Both auxin and cytokinin promote cell division which shows their synergistic effect on cell division.

(2) **Abscission :** Both ethylene and ABA are responsible for promoting the shedding of leaves, fruits and flowers.

Antagonistic effects

(1) **Dormancy :** ABA induces dormancy of buds, seeds and storage organs whereas gibberellins inhibit it.

(2) **Apical dominance :** Auxin promotes the apical dominance whereas it is inhibited by cytokinin. Thus, these two hormones act antagonistically.

(3) **Senescence :** Senescence is prevented by auxins while it is stimulated by ABA.

The role of PGR is of only one kind of intrinsic control. Along with genomic control and extrinsic factors, they play an important role in the plant growth and development. Many of the extrinsic factors such as temperature and light, control plant growth and development via PGR. Some of such events could be dormancy, seed germination, flowering, plant movements etc. Let us try to understand some of these.

Seed Dormancy

The seed undergoes a period of suspended growth and does not germinate as soon as it is formed. The suspension of growth is referred to as **quiescence** when it is due to exogenous factors, such as the environmental conditions. The seeds may be in a state of **dormancy** or rest due to endogenous control during which metabolic activity of the seed is greatly reduced. Quiescence is the condition of a seed when it is unable to germinate because the conditions for germination are not available. Such seeds will germinate if they are supplied with water and suitable temperature. While dormancy is the condition of seed when it is unable to germinate inspite of the availability of all environments conditions suitable for germination.

Factors : Dormancy in seed may be due to impermeable or mechanically resistant seed coats, rudimentary or physiologically immature embryos and even due to the presence of germination inhibitors such as abscisic acid, phenolic acid, short chain fatty acids and coumarin.

Breakage of dormancy : Dormancy of the seed can be broken, or its duration can be reduced to initiate germination, by mechanical or chemical scarification of the seed coat, stratification of seeds or changing environmental conditions such as temperature, light and pressure.

Scarification of seed involves scratching of seed coat to break the seed dormancy caused by hard and impermeable seed coat. Stratification of seed is subjecting the moist seeds to oxygen for variable periods of low or high temperatures.

Germination of seed / Seed germination

The resumption of active growth of the embryo present in the seed after a period of dormancy is known as germination. For the germination of a seed, water, temperature and oxygen are essential requirements. When a seed is provided with these essential requirements, it starts to germinate. During this process, first structure developed within the seed is known as radicle. Radicle breaks the seed coat and passes downwardly in the soil to establish itself as the primary root. Later it gives rise to tap root system. Soon after the elongation of the radicle, either hypocotyl or epicotyl shows active growth which pushes plumule out of the soil which form shoot system. The growth of radical and plumule is due to the cell extension, division and initiation of several biochemical processes. The seed also needs a suitable temperature (optimum between 25°C to 35°C). Rate of respiration increases rapidly during germination.

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In bean, hypocotyl grows actively and becomes curved. It brings the seed above the soil. After coming above the surface of the soil, the hypocotyl straightens. The loosened seed coats fall down. Now, epicotyl grows and the plumule gives rise to green leaves.

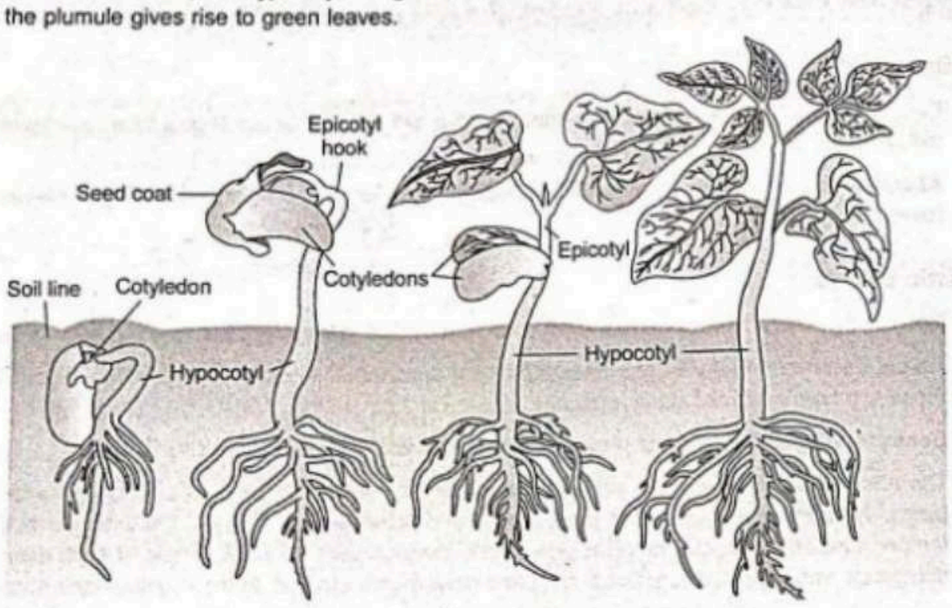


Fig. : Germination and seedling development in bean

Emergence of radicle by rupturing seed coat is called seed germination. It is of two types :

(a) **Epigeal :** Hypocotyl grows first, cotyledons comes out of soil, as in cucurbits, mustard, castor, onion, tamarind, bean etc.

(b) **Hypogeal :** Epicotyl grows first, cotyledons remains underground as in rice, maize, mango, gram, groundnut etc.

Vivipary

Vivipary is the germination of a seed while it is still attached to the parent plant and is nourished by it. It occurs in mangrove plants like *Rhizophora* and *Sonneratia*. These plants grow in saline marshy habitats along sea shores. Seeds cannot germinate in such habitats due to excessive salt and deficiency of oxygen. Mangroves plants have solved this problem through vivipary. The seed does not undergo dormancy, soon it germinates while present inside the fruit attached to parent plant. As the germinating seed forms a seedling, its weight increases and the seedling separates and falls down into the mud. The lateral roots then develop from radicle to help proper anchorage of the seedling.

Flowering

Plants flower when they reach the reproductive phase. There are various parameters which influence the process of flowering. Two main phenomena involved in the flowering process are photoperiodism and vernalisation.

PHOTOPERIODISM

The relative lengths of dark and light periods vary form place to place and from season to season which influence the flowering process to a great extent in different plants. The length of light period is called **photoperiod** and the responses shown by the plants to changes in the relative lengths of dark and light period is called **photoperiodism**. Thus, plants exhibiting such response are known as **photoperiodic plants**.

Garner and **Allard**, found that the tobacco plants could flower only if the plants were exposed to a number of short days. In autumn, this occurred naturally. But in summers the plants did not flower even after attaining much heights, until and unless these were provided a number of short days of 7 hours in a greenhouse. When they examined more plants for this behaviour, they found that plants differ in their requirements for day length.

Most plants can flower only if they were exposed to light for less or more than a certain period which is known as **critical period**. Accordingly, plants can be categorised as follows :

(1) **Long day plants :** These plants flower when they receive long photoperiods or light hours which are above than a critical period. e.g., wheat, radish, sugarbeet, henbane etc.

(2) **Short day plants :** These plants flower when they are exposed to a photoperiod shorter than a critical period. Most of winter-flowering plants belong to this category. e.g., soyabean, tobacco etc.

(3) **Day neutral plants :** These plants do not show any correlation between exposure to light duration and induction of flowering response. They can blossom throughout the year e.g., tomato, pepper, cucumber etc.

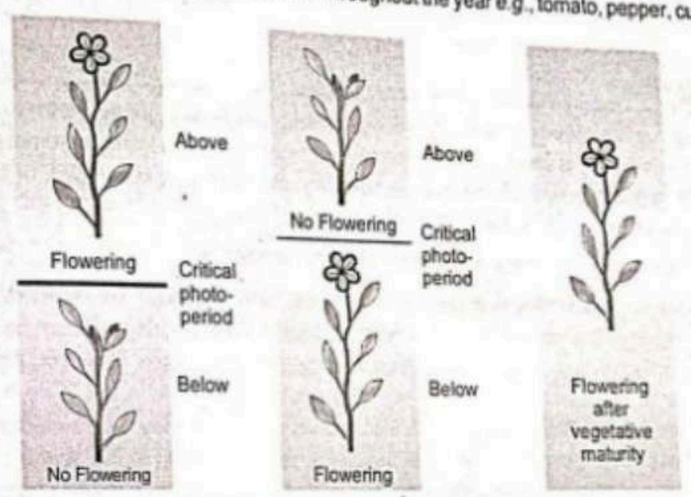


Fig. : Photoperiodism : Long day, short day and day neutral plants.

It is also interesting to note that before flowering, shoot apices modify themselves into flowering apices. Shoot apices of plants themselves cannot perceive photoperiods. **The site of perception of light/dark is the leaves of plants.** It has been hypothesised that there is a hormonal substance known as **florigen** which is responsible for flowering. When plants are exposed to the required inductive photoperiod, the hormonal substance *i.e.*, florigen migrates from leaves to shoot apices for inducing flowering.

Importance of Photoperiodism

(1) The knowledge of photoperiodic effect is useful to the commercial flower growers. It enables them to induce or retard flowering by regulating the photoperiodic and temperature conditions in glass houses to meet the demands of the market.

(2) The knowledge of photoperiodism is also useful in laying out gardens, orchards and planning crop pattern in a particular area.

(3) The knowledge of photoperiodism can be utilized in keeping some plants in vegetative growth to obtain higher yield of tubers, rhizomes etc., or keep the plant in reproductive stage to yield more flowers and fruits.

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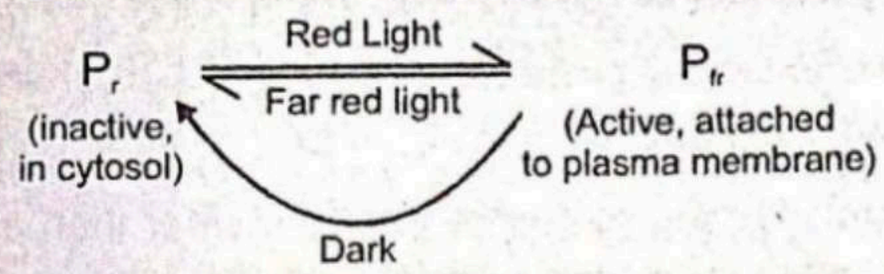
Note :

***Phytochrome** is a photoreceptor, a pigment that plants use to detect light. Important plant responses regulated by it are photoperiodic induction of flowering and light induced seed germination.*

Photoperiodic stimulus is believed to be perceived by phytochrome. It is a blue-green pigment existing in two interconvertible forms : P_{fr} or P_{730} (absorbs far-red lights) and P_r or P_{660} (absorb red light). By absorbing red light, P_r is converted to P_{fr} rapidly. P_{fr} on absorbing far-red light is converted to P_r rapidly. Darkness also causes this conversion.

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P_{fr} is physiologically active form; P_r is inactive.



For SDP, $\frac{P_r}{P_{fr}} > 1$ while for LDP, $\frac{P_{fr}}{P_r} > 1$ is critical for flowering.

VERNALISATION

There are various plants for which flowering is either quantitatively or qualitatively dependent on exposure to low temperature. This phenomenon is termed **vernalisation** or it refers to promotion of flowering by a period of low temperature. Here, the stimulus is perceived by the mature stem apex, or by the embryo of the seed, but not by the leaves as in photoperiodism.

Vernalisation is seen in many winter annuals and biennial plants

(a) **Winter annuals :** Some important food plants, wheat, barley, rye have two kinds of varieties– winter and spring. The 'spring' variety is normally planted in the spring season. In this variety, flowering and production of grain occurs before the end of the growing season. However, if winter varieties are planted earlier to their sowing period *i.e.*, in spring season, this would result in no flowering and production of mature grains within a span of a flowering season. Hence, they are planted in autumn. They germinate and over winter come out as small seedlings, resume growth in the spring, and are harvested usually around mid-summer. Therefore, we can say that plants require a specific temperature.

(b) **Biennial plants:** Biennial plants such as sugarbeet, cabbage, carrot are monocarpic that normally flower and die in the second season. When growing biennial plants are subjected to a cold treatment, it leads to the stimulation of a subsequent photoperiodic response which results in flowering. The main role of vernalisation is to prevent the precocious *i.e.*, early reproductive development so that the plants have enough time to reach maturity.

Importance of Vernalisation

(1) Vernalisation can help in shortening the period between germination and flowering. Thus, more than one crop can be obtained during a year.

(2) Sowing of winter crops (cereals) in spring season after vernalisation can avoid killing of cereals in severe winter.

Scan the QR code for further explanation

EXERCISE

11. Viviparous type of seed germination is found in
(1) *Sonneratia* (2) *Rhizophora*
(3) *Oryza sativa* (4) Both (1) & (2)

12. What is meant by vivipary in plants?
(1) Seed germination with subterranean cotyledons
(2) Seed germination with epiterranean cotyledons
(3) Fruit development without pollination
(4) Seed germination inside fruit, while the fruit is still attached to plant

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13. Phytochrome is responsible for
(1) Flowering
(2) Seed germination
(3) Transpiration
(4) Both (1) & (2)

14. Which of the following plants do not show any correlation between exposure to light duration and induction of flowering response?
(1) Long day plant
(2) Day neutral plant
(3) Short day plant
(4) All of these

15. Site of perception of light/dark duration is/are
(1) Shoot apex
(2) Root apex
(3) Permanent tissue
(4) More than one option is correct

16. Which of the following pair is **incorrectly** matched?
(1) Tomato – Day neutral plant
(2) Wheat – Short day plant
(3) Henbane – Long day plant
(4) Sugarcane – Short day plant

17. What is vernalisation?
(1) Flowering is induced only when plant is exposed to certain duration of photoperiod
(2) Flowering is either quantitatively or qualitatively dependent on exposure to low temperature
(3) Presence of different kinds of structure in response to environment or phases of life
(4) All changes that an organism goes through during its life cycle

18. Stimulus of low temperature in vernalisation is perceived by
(1) Leaves
(2) Embryo of the seed
(3) Flower
(4) Pericarp

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Assignment

SECTION - A

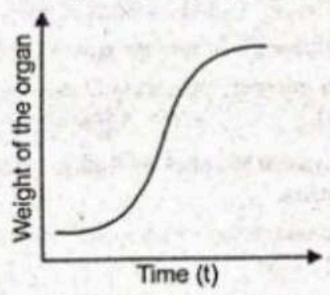
NCERT Based MCQs

1. Which of the following pairs is **not** correctly matched? **[NCERT Pg. 247-250]**
   (1) IAA : *Avena* curvature bioassay
   (2) ABA : Stomatal closure
   (3) Cytokinin : Cell division
   (4) Gibberellic acid : Stress hormone
2. Select the **odd** one w.r.t. functions of auxin **[NCERT Pg. 248]**
   (1) Induces rooting
   (2) Apical dominance
   (3) To speed up the malting process
   (4) Phototropism
3. In a plant, ethylene **[NCERT Pg. 250]**
   (1) Is a gaseous hormone
   (2) Causes dormancy
   (3) Is a waste product
   (4) Causes Bakanae disease
4. Plant growth differs from animals as it is **[NCERT Pg. 240]**
   (1) Indeterminate
   (2) Open form of growth
   (3) Definite and diffused
   (4) Both (1) & (2)
5. All of the following are formed from dedifferentiation of parenchyma cells, **except** **[NCERT Pg. 245]**
   (1) Wound cambium
   (2) Cork cambium
   (3) Interfascicular cambium
   (4) Intrafascicular cambium
6. The plant hormone which is a derivative of carotenoids is **[NCERT Pg. 247]**
   (1) ABA
   (2) IAA
   (3) $CH_2 = CH_2$
   (4) Cytokinin
7. Leaf 'A' of initial size 10 cm^2 grows 5 cm^2 per year and leaf 'B' of 25 cm^2 grows 5 cm^2 per year.
   The relative growth rates of leaf A & B respectively are **[NCERT Pg. 244]**
   (1) 50% and 25% (2) 50% and 20%
   (3) 10% and 20% (4) 20% and 25%
8. If the apical bud is removed from a plant then what will happen to it? **[NCERT Pg. 248]**
   (1) It will show yellowing of leaves
   (2) Its growth will stop
   (3) It will stop flowering & becomes deficient in zinc
   (4) It will show more lateral branches & axillary buds are activated
9. A phytohormone which was first isolated from a fungus **[NCERT Pg. 249]**
   (1) Is basic in nature
   (2) Helps in seed germination
   (3) Decreases sugarcane yield
   (4) Is used to promote fruit ripening
10. Choose the **odd** one w.r.t. heterophylly controlled by intrinsic factors. **[NCERT Pg. 246]**
    (1) Cotton (2) Coriander
    (3) Buttercup (4) Larkspur
11. A synthetic auxin **'A'** is widely used as weedicide in cereal crops.
    Choose the **correct** option for 'A'. **[NCERT Pg. 249]**
    (1) IBA (2) IAA
    (3) 2, 4-D (4) NAA
12. Treatment of a plant with cytokinin enhances **[NCERT Pg. 250]**
    (1) Chloroplast production in leaves
    (2) Initiation of growth of apical buds
    (3) Ageing of leaves
    (4) Parthenocarpy in all plants
13. Coconut milk contains **[NCERT Pg. 249]**
    (1) Ethylene (2) ABA
    (3) Cytokinin (4) Gibberellin

14. The term antagonistic action of hormones refers to the condition when **[NCERT Pg. 250]**
    (1) Two hormones act together and contribute the same function
    (2) Many hormones bring about any one function
    (3) Two hormones bring about opposite effects on one function
    (4) One hormone affects more than one function
15. ABA is responsible for promoting **[NCERT Pg. 250]**
    (1) Stem elongation
    (2) Cell elongation
    (3) Abscission of flower and dormancy of buds
    (4) Root elongation
16. If banana fruits are plucked unripe from the plant then which hormone should be used for ripening of banana fruits? **[NCERT Pg. 250]**
    (1) Cytokinin (2) Auxin
    (3) ABA (4) Ethylene
17. One of the plant hormones which gets removed along with apical bud of tea plants is **[NCERT Pg. 248]**
    (1) A gaseous hormone
    (2) The hormone first isolated from human urine
    (3) Called kinetin
    (4) An adenine derivative
18. Typical growth curve in plants is **[NCERT Pg. 243]**
    (1) Parabolic (2) Hyperbolic
    (3) Linear (4) Sigmoid
19. The phytohormone which can replace vernalisation in some plants is **[NCERT Pg. 253]**
    (1) Ethylene (2) ABA
    (3) Gibberellin (4) Auxin
20. Zygotic divisions initially **[NCERT Pg. 242]**
    (1) Show arithmetic growth
    (2) Can be expressed by a straight line
    (3) Show a pattern where every cell divides including all the daughter cells
    (4) Has only one cell dividing whereas other cells undergo differentiation
21. Abscisic acid **[NCERT Pg. 250]**
    (1) Delays senescence in leaves and other plant parts
    (2) Generally acts as antagonist of gibberellin
    (3) Is the major component of culture medium for proliferation of callus
    (4) Promotes rooting and root hair formation
22. Identify the **incorrectly** matched pair. **[NCERT Pg. 248-250]**
    (1) Auxin – Phototropism
    (2) Gibberellins – Foolish seedling disease of rice
    (3) Ethylene – Induces fruit ripening
    (4) Cytokinins – Promote apical dominance
23. Find the **correct** application of cytokinin. **[NCERT Pg. 248-250]**
    (1) Acts as antitranspirant and induces closure of stomata
    (2) To promote root development on various types of cuttings
    (3) In tea plantation for giving bushy appearance
    (4) Promotes parthenocarpy in fruits like banana and grapes
24. Increased vacuolation is a feature of cells of which of the following growth phase in plants? **[NCERT Pg. 242]**
    (1) Phase of cell division
    (2) Phase of elongation
    (3) Phase of differentiation
    (4) Phase of maturation
25. Which of the following plant hormones is not acidic in nature? **[NCERT Pg. 247]**
    (1) Gibberellin (2) Auxin
    (3) Cytokinin (4) ABA
26. Auxin promotes the apical dominance whereas it is counteracted by _____.
    Complete the statement by choosing the **correct** option. **[NCERT Pg. 250]**
    (1) Gibberellin (2) Cytokinin
    (3) Ethylene (4) ABA
27. Which phytohormone stimulates cell division and delays senescence? **[NCERT Pg. 249-250]**
    (1) Auxins
    (2) Gibberellins
    (3) Cytokinins
    (4) Vernalins
28. Growth is measured by all of the following parameters, **except** **[NCERT Pg. 241]**
    (1) Increase in fresh weight
    (2) Increase in cell number
    (3) Increase in life-span
    (4) Increase in cell size

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29. If W_0 and W_1 are initial and final weight of a plant organ and r and e are growth rate and base of natural logarithms respectively, then, which of the following equations will satisfy the growth of that organ represented by the given graph? **[NCERT Pg. 243]**



(1) $W_1 = W_0 e^{rt}$
(2) $W_1 = W_0 / e^{rt}$
(3) $W_1 = W_0 + e^{rt}$
(4) $W_1 = W_0 . r/e^t$

30. The discovery of gibberellin is related to which of the following plant diseases? **[NCERT Pg. 248]**
(1) White rust of mustard
(2) Bakane disease of rice seedlings
(3) Mosaic disease of tobacco
(4) Early blight of potato

31. Which of the following **cannot** be a cause of seed dormancy? **[NCERT Pg. 252]**
(1) Absence of endosperm
(2) Impermeability of seed coat
(3) Presence of abscisic acid
(4) Immaturity of embryo

32. Select the **incorrect** match from the following. **[NCERT Pg. 249]**
(1) 2, 4-D – Used to kill dicotyledonous weeds
(2) Gibberellins – Promote bolting in beet
(3) Cytokinins – Promote abscission in mature fruits
(4) Ethylene – Induces flowering in mango

33. The site of perception of light/dark duration that induces flowering in short day or long day plants are **[NCERT Pg. 252]**
(1) Leaves
(2) Shoot apices
(3) Internodes
(4) Buds

34. Vernalisation prevents the plant from **[NCERT Pg. 252]**
(1) Intense heat of the environment
(2) Disease caused by viruses
(3) High intensity of light during summer
(4) Precocious reproductive development late in the growing season

35. Which of the following hormone is responsible for phototropic movements in coleoptile tip of canary grass? **[NCERT Pg. 247]**
(1) Auxin (2) Gibberellins
(3) ABA (4) Ethylene

36. Most intensively studied form of gibberellins is **[NCERT Pg. 249]**
(1) GA_1 (2) GA_2
(3) GA_3 (4) GA_4

37. Phytohormones are composed of different types of chemical compounds. Match the following columns and choose the **correct** option. **[NCERT Pg. 247]**

Column I (Composition)	Column II (Phytohormone)
a. Indole compounds	(i) Cytokinins
b. Terpenes	(ii) Auxins
c. Gas	(iii) Gibberellins
d. Adenine derivatives	(iv) Ethylene

(1) a(ii), b(iii), c(iv), d(i) (2) a(iii), b(ii), c(iv), d(i)
(3) a(ii), b(iii), c(i), d(iv) (4) a(ii), b(i), c(iii), d(iv)

38. Which of the following is **not** a function of ethylene? **[NCERT Pg. 250]**
(1) Sprouting of potato tuber
(2) Promotes rapid elongation of internode in deep water rice plant
(3) Thinning of cotton fruits
(4) Promotes male flowers in cucumber

39. Promotion of flowering by a period of low temperature is called **[NCERT Pg. 252]**
(1) Thermotropism
(2) Vernalisation
(3) Photoperiodism
(4) Abscission

40. Arithmetic growth of plants is mathematically expressed by the given equation
$L_t = L_o + rt$
What does 'r' indicates in this equation. **[NCERT Pg. 243]**
(1) Growth rate during log phase
(2) Length of organ at time "zero"
(3) Total growth per unit time
(4) Elongation per unit time

SECTION - B

Objective Type Questions

1. Which of the following phytohormone is responsible for inducing horizontal growth of seedlings?
(1) Ethylene (2) ABA
(3) Auxin (4) Gibberlin

2. Environmental plasticity is shown by
(1) Cotton (2) Buttercup
(3) Coriander (4) Larkspur

3. Match column I with column II and choose the correct option

Column I	Column II
a. Tomato	(i) Vivipary
b. P_{fr}	(ii) Bioassay of ethylene
c. *Rhizophora*	(iii) Physiologically active form
d. Triple response test	(iv) DNP

(1) a(ii), b(iii), c(iv), d(i)
(2) a(iv), b(iii), c(i), d(ii)
(3) a(iii), b(ii), c(iv), d(i)
(4) a(iv), b(ii), c(iii), d(i)

4. Chilling treatment prior to flowering
(1) Is required for spring varieties of cereals
(2) Requires a temperature treatment between 0°C to 5°C to the mature tissues of plant
(3) Hastens reproductive maturity early in the growing season
(4) Is essential for both biennials and perennials

5. Select the **incorrect** match
(1) IAA – Terpenes
(2) C_2H_4 – Gaseous hydrocarbon which also function as PGR
(3) Cytokinin – Benzyl amino purine
(4) ABA – Stress hormone

6. The photoperiodic stimulus perceived by __(A)__ migrates to __(B)__ for producing flowers
(1) A - Roots, B - Bud
(2) A - Meristem, B - Leaves
(3) A - Leaves, B - Shoot apex
(4) A - Bud, B - Leaves

7. A plant is subjected to alternating periods of 2 hrs light and 6 hrs dark, in a 24 hrs cycle for flowering what is the type of plant described above?
(1) SDP (2) LDP
(3) DNP (4) L-SDP

8. Choose the **odd** one w.r.t. examples of epigeal germination
(1) Maize (2) Mustard
(3) Castor (4) Cucurbits

9. Match the columns w.r.t. precursor of PGRs

Column-I	Column-II
a. ABA	(i) Acetyl CoA
b. IAA	(ii) Methionine
c. GA	(iii) Tryptophan
d. Ethylene	(iv) Violaxanthin

(1) a(iv), b(iii), c(i), d(ii)
(2) a(i), b(iii), c(ii), d(iv)
(3) a(i), b(iv), c(iii), d(ii)
(4) a(iv), b(ii), c(iii), d(i)

10. Thinning of cotton is caused by
(1) Ethylene
(2) Cytokinins
(3) Gibberellin
(4) IAA

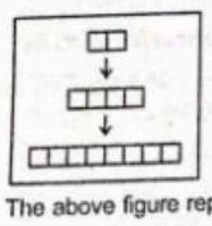
11. A – Gibberellin promotes bolting in beet and cabbages just after flowering.
B – Kinetin does not occur naturally in plants.
(1) Both are correct
(2) Only A is incorrect
(3) Only B is incorrect
(4) Both are incorrect

12. Which of the following hormones is antagonistic to a PGR of terpene nature/composition in most of the cases?
(1) ABA (2) Zeatin
(3) Ethylene (4) Gibberellin

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13. Choose the **correct** statement w.r.t. the hormone found in all group of plants and discovered in fungus.
(1) It hastens the maturity period in conifers leading to early seed production
(2) It promotes flowering in pineapples
(3) Controls xylem differentiation
(4) Promotes root growth and root hair formation

14. Choose the **Incorrect** statement w.r.t. plant growth.
(1) Plant growth is indeterminate
(2) It is localised
(3) Juvenile stage is similar to adult stage w.r.t. the number of organs
(4) Plant growth is measured with the help of auxanometer

15. 
The above figure represents
(1) Arithmetic growth as shown during differentiation of embryo
(2) Geometric growth as shown during morphogenesis and differentiation in embryo
(3) Geometric growth as shown during initial phase of embryogenesis
(4) It will be represented by a straight line when plotted against time

16. Developmental phenotypic plasticity is seen in
(1) Buttercup, Larkspur
(2) Butter cup, cotton
(3) Cotton, coriander
(4) Amphibious plants only

17. Read the given statements and select the correct option
(A) Nutrients are not required for growth
(B) Turgidity of cells helps in extension growth
(C) Environmental factors affect different stages of growth
(1) All statements are correct
(2) Only (B) is incorrect
(3) Only (A) is incorrect
(4) Both (B) and (C) are incorrect

18. Which of the following is an example of redifferentiation?
(1) Secondary cortex formation
(2) Deposition of pectin non-uniformly in corners of collenchyma
(3) Formation of cork cambium form cortical cell in dicot stem
(4) Deposition of cutin to form cuticle in leaf

19. Tissue culture technique which involves formation of callus represents
(1) Growth and differentiation
(2) Growth only
(3) Dedifferentiation
(4) Differentiation only

20. Select the **incorrect** match
(1) Herring sperm DNA - N_6-furfurylaminopurine
(2) Stomatal closure - Carotenoids derived hormone
(3) Weed free lawn - Synthetic auxin
(4) Bakanae disease - Gaseous hydrocarbon

21. The stimulus of vernalisation is perceived by
(1) Roots
(2) Shoot apices
(3) Mature leaf
(4) Flowers

22. Ethephon is **not** involved in
(1) Hastening of ripening of fruits in tomatoes and apples
(2) Thinning of cotton, cherry, walnut
(3) Femaleness in cucumber
(4) Nutrient mobilisation which helps in the delay of leaf senescence

23. Which of the following functions is **correct** for a phytohormone whose discovery was based as a result of three independent researches?
(1) Promotes growth of plant
(2) Initiates sprouting of potato tuber
(3) Plays an important role in seed development, maturation and dormancy
(4) Hastens fruit ripening in tomatoes

24. Choose the **odd** one w.r.t. the examples showing antagonistic effects of PGR.
(1) Auxin and cytokinin w.r.t. apical dominance
(2) Gibberellin and ABA w.r.t. seed germination
(3) Auxin and ethylene w.r.t. flowering in pineapples
(4) Ethylene and gibberellin w.r.t. sex expression

25.

The above figure shows flowering in a particular type of plant. Choose the **correct** statement for this plant.
(1) The plant flowers when $\frac{P_{FR}}{P_R} < 1$
(2) It is a day neutral plant
(3) The plant always has a critical photoperiod more than 12 hrs
(4) Henbane shows flowering under such condition

26. Vernalisation
(1) Is only quantitative dependence on low temperature for flowering in certain plants
(2) Is shown by spring varieties of cereals like rye, barley and wheat
(3) Stimulates a subsequent photoperiodic flowering response in biennials
(4) It promotes precocious reproductive development late in the growing season

27. Seed dormancy is regulated by
(1) PGR which acts as a general plant growth inhibitor
(2) GA_3 which inhibits formation of α-amylase in seeds
(3) Cytokinin which promotes cell division
(4) Auxin which promotes differentiation

28. Vernalisation can be replaced by
(1) Gibberellin treatment
(2) Auxin treatment
(3) Cytokinin treatment
(4) Ethylene treatment

29. Development is controlled by
(1) PGR only
(2) Environmental, genetic and intercellular factors
(3) Intracellular factors only
(4) Intrinsic factors only

30. Phytohormone which plays a very important role in brewing industry also
(1) Causes root initiation in plants
(2) Initiates production of new leaves and chloroplast
(3) Causes seed development and dormancy
(4) Leads to early seed production in conifers

SECTION - C
Previous Years Questions

1. The ability of plants to follow different pathways in response to environment leading to formation of different kinds of structures is called **[NEET-2022 (Phase-2)]**
(1) Differentiation (2) Redifferentiation
(3) Development (4) Plasticity

2. Which of the following growth regulators is an adenine derivative? **[NEET-2022 (Phase-2)]**
(1) Abscisic acid (2) Auxin
(3) Cytokinin (4) Ethylene

3. The Phenomenon by which the undividing parenchyma cells start to divide mitotically during plant tissue culture is called as **[NEET-2022 (Phase-2)]**
(1) Secondary growth (2) Differentiation
(3) Dedifferentiation (4) Redifferentiation

4. Which one of the following plants does not show plasticity? **[NEET-2022 (Phase-1)]**
(1) Buttercup (2) Maize
(3) Cotton (4) Coriander

5. Production of Cucumber has increased manifold in recent years. Application of which of the following phytohormones has resulted in this increased yield as the hormone is known to produce female flowers in the plants : **[NEET-2022 (Phase-1)]**
(1) Ethylene (2) Cytokinin
(3) ABA (4) Gibberellin

6. The gaseous plant growth regulator is used in plants to : **[NEET-2022 (Phase-1)]**
(1) help overcome apical dominance
(2) kill dicotyledonous weeds in the fields
(3) speed up the malting process
(4) promote root growth and roothair formation to increase the absorption surface

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7. Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called **[NEET-2021]**
(1) Maturity (2) Elasticity
(3) Flexibility (4) Plasticity

8. The site of perception of light in plants during photoperiodism is **[NEET-2021]**
(1) Leaf (2) Shoot apex
(3) Stem (4) Axillary bud

9. The plant hormone used to destroy weeds in a field **[NEET-2021]**
(1) IBA (2) IAA
(3) NAA (4) 2, 4-D

10. Which of the following is **not** an inhibitory substance governing seed dormancy? **[NEET-2020 (Phase-1)]**
(1) Abscisic acid (2) Phenolic acid
(3) Para-ascorbic acid (4) Gibberellic acid

11. Name the plant growth regulator which upon spraying on sugarcane crop, increases the length of stem, thus increasing the yield of sugarcane crop. **[NEET-2020 (Phase-1)]**
(1) Gibberellin (2) Ethylene
(3) Abscisic acid (4) Cytokinin

12. The process of growth is maximum during **[NEET-2020 (Phase-1)]**
(1) Lag phase (2) Senescence
(3) Dormancy (4) Log phase

13. Inhibitory substances in dormant seeds cannot be removed by subjecting seeds to: **[NEET-2020 (Phase-2)]**
(1) Chilling conditions (2) Gibberellic acid
(3) Nitrate (4) Ascorbic acid

14. Match the following concerning the activity/function and the phytohormone involved. **[NEET-2020 (Phase-2)]**

(a)	Fruit ripener	(i)	Abscisic acid
(b)	Herbicide	(ii)	GA_3
(c)	Bolting agent	(iii)	2, 4-D
(d)	Stress hormone	(iv)	Ethephon

Select the correct option from following
(1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

15. Who coined the term 'Kinetin'? **[NEET-2020 (Phase-2)]**
(1) Kurosawa (2) Skoog and Miller
(3) Darwin (4) Went

16. Removal of shoot tips is a very useful technique to boost the production of tea-leaves. This is because: **[NEET-2019 (Odisha)]**
(1) Gibberellins delay senescence of leaves.
(2) Gibberellins prevent bolting and are inactivated.
(3) Auxins prevent leaf drop at early stages.
(4) Effect of auxins is removed and growth of lateral buds is enhanced.

17. What is the site of perception of photoperiod necessary for induction of flowering in plants? **[NEET-2019]**
(1) Lateral buds (2) Pulvinus
(3) Shoot apex (4) Leaves

18. It takes very long time for pineapple plants to produce flowers. Which combination of hormones can be applied to artificially induce flowering in pineapple plants throughout the year to increase yield? **[NEET-2019]**
(1) Auxin and Ethylene
(2) Gibberellin and Cytokinin
(3) Gibberellin and Abscisic acid
(4) Cytokinin and Abscisic acid

19. Fruit and leaf drop at early stages can be prevented by the application of **[NEET-2017]**
(1) Cytokinins (2) Ethylene
(3) Auxins (4) Gibberellic acid

20. You are given a tissue with its potential for differentiation in an artificial culture. Which of the following pairs of hormones would you add to the medium to secure shoots as well as roots? **[NEET-2016 (Phase-2)]**
(1) IAA and gibberellin
(2) Auxin and cytokinin
(3) Auxin and abscisic acid
(4) Gibberellin and abscisic acid

21. Phytochrome is a **[NEET-2016 (Phase-2)]**
(1) Flavoprotein (2) Glycoprotein
(3) Lipoprotein (4) Chromoprotein

22. The avena curvature is used for bioassay of **[NEET-2016]**
(1) Ethylene (2) ABA
(3) GA_3 (4) IAA

23. Auxin can be bioassayed by **[Re-AIPMT-2015]**
(1) Lettuce hypocotyl elongation
(2) Avena coleoptile curvature
(3) Hydroponics
(4) Potometer

24. Typical growth curve in plants is **[AIPMT-2015]**
(1) Parabolic (2) Sigmoid
(3) Linear (4) Stair-steps shaped

25. What causes a green plant exposed to the light on only one side, to bend toward the source of light as it grows? **[AIPMT-2015]**
(1) Auxin accumulates on the shaded side, stimulating greater cell elongation there
(2) Green plants need light to perform photosynthesis
(3) Green plants seek light because they are phototropic
(4) Light stimulates plant cells on the lighted side to grow faster

26. A few normal seedings of tomato were kept in a dark room. After a few days they were found to have become white-coloured like albinos. Which of the following terms will you use to describe them? **[AIPMT-2014]**
(1) Mutated (2) Embolised
(3) Etiolated (4) Defoliated

27. Which one of the following growth regulators is known as stress hormone? **[AIPMT-2014]**
(1) Abscissic acid (2) Ethylene
(3) GA_3 (4) Indole acetic acid

28. Dr.F. Went noted that if coleoptile tips were removed and placed on agar for one hour, the agar would produce a bending when placed on one side of freshly - cut coleoptile stumps. Of what significance is this experiment? **[AIPMT-2014]**
(1) It made possible the isolation and exact identification of auxin
(2) It is the basis for quantitative determination of small amounts of growth-promoting substances
(3) It supports the hypothesis that IAA is auxin
(4) It demonstrated polar movement of auxins

29. During seed germination its stored food is mobilized by **[NEET-2013]**
(1) Cytokinin
(2) ABA
(3) Gibberellin
(4) Ethylene

30. Through their effect on plant growth regulators, what do the temperature and light control in the plants? **[AIPMT (Mains)-2012]**
(1) Apical dominance (2) Flowering
(3) Closure of stomata (4) Fruit elongation

31. Which one of the following generally acts as an antagonist to gibberellins? **[AIPMT (Mains)-2012]**
(1) Zeatin (2) Ethylene
(3) ABA (4) IAA

32. Vernalisation stimulates flowering in **[AIPMT (Mains)-2012]**
(1) Zamikand (2) Turmeric
(3) Carrot (4) Ginger

33. Photoperiodism was first characterised in **[AIPMT (Prelims)-2010]**
(1) Cotton (2) Tobacco
(3) Potato (4) Tomato

34. Phototropic curvature is the result of uneven distribution of **[AIPMT (Prelims)-2010]**
(1) Auxin (2) Gibberellin
(3) Phytochrome (4) Cytokinins

35. Coiling of garden pea tendrils around any support is an example of **[AIPMT (Prelims)-2010]**
(1) Thermotaxis (2) Thigmotaxis
(3) Thigmonasty (4) Thigmotropism

36. One of the commonly used plant growth hormone is tea plantations is **[AIPMT (Mains)-2010]**
(1) Ethylene (2) Abscisic acid
(3) Zeatin (4) Indole-3-acetic acid

37. Root development is promoted by **[AIPMT (Mains)-2010]**
(1) Abscisic acid (2) Auxin
(3) Gibberellin (4) Ethylene

38. One of the synthetic auxin is **[AIPMT (Prelims)-2009]**
(1) IAA (2) GA
(3) IBA (4) NAA

39. Which one of the following acids is a derivative of carotenoids? **[AIPMT (Prelims)-2009]**
(1) Indole-3-acetic acid
(2) Gibberellic acid
(3) Abscisic acid
(4) Indole butyric acid

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40. Senescence as an active developmental cellular process in the growth and functioning of a flowering plant, is indicated in **[AIPMT (Prelims)-2008]**
(1) Floral parts
(2) Vessels and tracheid differentiation
(3) Leaf abscission
(4) Annual plants

41. Importance of day length in flowering of plants was first shown in **[AIPMT (Prelims)-2008]**
(1) Petunia (2) Lemna
(3) Tobacco (4) Cotton

42. "Foolish Seedling" disease of rice led to the discovery of **[AIPMT (Prelims)-2007]**
(1) IAA (2) GA
(3) ABA (4) 2, 4 – D

43. Opening of floral buds into flowers, is type of: **[AIPMT (Prelims)-2007]**
(1) Autonomic movement of growth
(2) Autonomic movement of locomotion
(3) Autonomic movement of variation
(4) Paratonic movement of growth

44. Which one of the following pairs, is **not correctly** matched? **[AIPMT (Prelims)-2007]**
(1) IAA – Cell wall elongation
(2) Abscissic acid – Stomatal closure
(3) Gibberellic acid – Leaf fall
(4) Cytokinin – Cell division

45. The wavelength of light absorbed by Pr form of phytochrome is **[AIPMT (Prelims)-2007]**
(1) 660 nm (2) 720 nm
(3) 620 nm (4) 640 nm

46. How does pruning help in making the hedge dense? **[AIPMT (Prelims)-2006]**
(1) It induces the differentiation of new shoots from the rootstock
(2) It frees axillary buds from apical dominance
(3) The apical shoot grows faster after pruning
(4) It releases wound hormones

47. Treatment of seed at low temperature under moist conditions to break its dormancy is called : **[AIPMT (Prelims)-2006]**
(1) Scarification (2) Vernalization
(3) Chelation (4) Stratification

48. An enzyme that can stimulate germination of barley seeds is **[AIPMT (Prelims)-2006]**
(1) α-amylase (2) Lipase
(3) Protease (4) Invertase

49. Farmers in a particular region were concerned that pre-mature yellowing of leaves of a pulse crop might cause decrease in the yield. Which treatment could be most beneficial to obtain maximum seed yield? **[AIPMT (Prelims)-2006]**
(1) Application of iron and magnesium to promote synthesis of chlorophyll
(2) Frequent irrigation of the crop
(3) Treatment of the plants with cytokinins along with a small dose of nitrogenous fertilizer
(4) Removal of all yellow leaves and spraying the remaining green leaves with 2, 4, 5-trichlorophenoxy acetic acid

50. The ability of the venus flytrap to capture insects is due to : **[AIPMT (Prelims)-2005]**
(1) Chemical stimulation by the prey
(2) A passive process requiring no special ability on the part of the plant
(3) Specialized "muscle-like" cells
(4) Rapid turgor pressure changes

Questions asked Prior to Medical Ent. Exams. 2005

51. The pineapple which under natural conditions is difficult to blossom has been made to produce fruits throughout the year by application of
(1) IAA, IBA
(2) NAA, 2, 4-D
(3) Phenyl acetic acid
(4) Cytokinin

52. If the growing plant is decapitated, then
(1) Its growth stops
(2) Leaves become yellow and fall down
(3) Axillary buds are inactivated
(4) Axillary buds are activated

53. 2, 4-D is an effective
(1) Rodenticide (2) Wormicide
(3) Fungicide (4) Weedicide

54. If the apical bud has been removed then we observe
(1) More lateral branches
(2) More adventitious buds
(3) Plant growth stops
(4) Flowering stops

55. Which of the following prevents the fall of fruits?
(1) GA_3 (2) NAA
(3) Ethylene (4) Zeatin

56. The maximum growth rate occurs in
(1) Stationary phase (2) Senescent phase
(3) Lag phase (4) Exponential phase

57. Gibberellic acid induces flower
(1) In short day plants under long day conditions
(2) In day-neutral plants under dark conditions
(3) In some gymnospermic plants only
(4) In long day plants under short day conditions

58. Which breaks dormancy of potato tuber?
(1) Gibberellin (2) IAA
(3) ABA (4) Zeatin

59. Cell elongation in internodal regions of the green plants takes place due to
(1) Indole acetic acid (2) Cytokinins
(3) Gibberellins (4) Ethylene

60. Natural cytokinins are synthesized in tissues that are
(1) Senescent (2) Dividing rapidly
(3) Storing food material (4) Differentiating

61. Differentiation of shoot is controlled by
(1) High auxin : cytokinin ratio
(2) High cytokinin : auxin ratio
(3) High gibberellin : auxin ratio
(4) High gibberellin : cytokinin ratio

62. Coconut milk factor is
(1) Auxin (2) A Gibberellin
(3) Abscissic acid (4) Cytokinin

63. Which hormone is responsible for fruit ripening?
(1) Ethylene (2) Auxin
(3) Ethyl chloride (4) Cytokinin

64. ABA is involved in
(1) Shoot elongation (2) Increased cell division
(3) Dormancy of seeds (4) Root elongation

65. Hormone responsible for senescence
(1) ABA (2) Auxin
(3) GA (4) Cytokinin

66. By which action a seed coat becomes permeable to water?
(1) Scarification (2) Stratification
(3) Vernalization (4) All of these

67. The response of different organisms to the environmental rhythms of light and darkness is called
(1) Vernalization (2) Photoperiodism
(3) Phototaxis (4) Phototropism

68. Which plant is LDP?
(1) Tobacco (2) *Glycine max*
(3) *Mirabilis jalapa* (4) Spinach

69. Proteinaceous pigment which control the activities concerned with light
(1) Phytochrome (2) Chlorophyll
(3) Anthocyanin (4) Carotenoids

70. One set of the plant was grown at 12 hours day and 12 hours night period cycles and it flowered while in the other set night phase was interrupted by flash of light and it did not produce flower. Under which one of the following categories will you place other set of plant?
(1) Long day (2) Darkness neutral
(3) Day neutral (4) Short day

71. What will be the effect on phytochrome in a plant subjected to continuous red light?
(1) Phytochrome synthesis will increase
(2) Level of phytochrome will decrease
(3) Phytochrome will be destroyed
(4) First (2) then (1)

72. The pigment, that absorbs red and far red light in plants, is
(1) Phytochrome (2) Carotene
(3) Xanthophyll (4) Cytochrome

73. Phytochrome becomes active in
(1) Red light (2) Green light
(3) Blue light (4) None of these

74. The twinning of tendrils around a support is a good example of
(1) Phototropism (2) Chemotropism
(3) Nastic movements (4) Thigmotropism

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75. The closing and opening of the leaves of *Mimosa pudica* is due to
(1) Seismonastic movement
(2) Chemonastic movement
(3) Thermonastic movement
(4) Hydrotropic movement

SECTION - D

NEET Booster Questions

1. The phenomenon of the inhibiting the growth of the lateral buds by growing apical bud is called
(1) Dormancy
(2) Apical dominance
(3) Bolting
(4) Triple response

2. All are the functions of adenine derivative phytohormone, **except**
(1) Promotes nutrient mobilisation
(2) Delays leaf senescence
(3) Accelerates thinning of cotton, cherry and walnut
(4) Promotes adventitious shoot formation

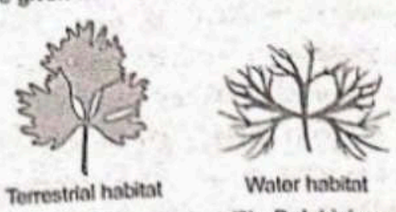
3. Choose the **mismatched** pair.
(1) Rooting hormone – Gibberellins
(2) Stress hormone – ABA
(3) Antiaging hormone – Cytokinin
(4) Ripening hormone – Ethylene

4. Which of the following is used to increase the market period of fruit and to kill dicot weeds respectively?
(1) GA_3, 2, 4-D (2) BAP, C_2H_4
(3) GA_3, BAP (4) C_2H_4, ABA

5. Pollen tube growth is measured in terms of
(1) Surface area (2) Cell number
(3) Fresh weight (4) Length

6. Elongation of root at constant rate is
(1) Arithmetic growth; S-curve; $L_0 = L_1 + rt$
(2) Arithmetic growth; Linear curve, $L_1 = L_0 + rt$
(3) Exponential growth; S-curve; $W_1 = W_0 e^{rt}$
(4) Exponential growth; Linear curve; $W_0 = W_1 e^{rt}$

7. The given heterophylly is present in



Terrestrial habitat Water habitat
(1) Butter cup (2) *Delphinium*
(3) Cotton (4) Coriander

8. In which of the following plants there is no correlation between exposure to light duration and induction of flowering response?
(1) Long day plant; Wheat
(2) Short day plant; Rice
(3) Long night plant; Tomato
(4) Day neutral plant; Maize

9. Find the **correct** option w.r.t. Vernalisation
(1) Induction of flowering by only quantitative exposure to low temperature
(2) Prevents precocious reproductive development late in the growing season and enables plant to reach maturity in sufficient time
(3) Site of perception is mature leaf
(4) Shown by annuals, biennials and perennials

10. Match the following

Column-I (PGR)	Column-II (Function)
a. Cytokinins	(i) Flowering in pineapple
b. Auxins	(ii) Lateral shoot growth
c. Ethylene	(iii) Flowering in mango
d. ABA	(iv) Seed development
	(v) Promotes root growth

(1) a(ii), b(i), c(iii), d(iv) (2) a(ii), b(iv), c(iii), d(v)
(3) a(iii), b(v), c(ii), d(iv) (4) a(i), b(iii), c(ii), d(v)

11. Which of the following can cause seed dormancy?
(1) Ethylene (2) Gibberellins
(3) Cytokinin (4) Abscisic acid

12. Find the **odd** one w.r.t. short day plants (SDP).
(1) Soyabean (2) Sugarcane
(3) Tobacco (4) Radish

13. Which of the following plant growth regulator plays an important role in seed development, maturation and dormancy?
(1) Ethylene (2) Abscisic acid
(3) Auxins (4) Cytokinins

14. Which of the following phytohormones help to prevent fruit and leaf drop at early stages but promote the abscission of older mature leaves and fruits?
(1) Auxins
(2) ABA
(3) Cytokinins
(4) Gibberellins

15. Consider the following statements w.r.t. gibberellins (GAs) and select right choice.
(a) More than 100 GAs are reported from fungi only
(b) All GAs are acidic
(c) Help in early seed production in conifers
(d) They also delay senescence
(1) Only (d) is correct
(2) (a), (b) & (d) are correct
(3) (b), (c) & (d) are correct
(4) (a) & (b) are correct

16. Cytokinins are synthesised in
(1) Root apices
(2) Developing shoot buds
(3) Young fruits
(4) More than one option is correct

17. Select the **incorrect** statement w.r.t. cells of meristematic zone during growth.
(1) The cells are rich in protoplasm
(2) Possess large conspicuous nucleus
(3) Cell walls are secondary in nature
(4) Cells have high karyoplasmic ratio

18. Which of the following is **wrong** statement w.r.t. ethylene?
(1) Promotes abscission of leaves
(2) Most widely used PGR in agriculture
(3) Inhibits the respiration during ripening of the fruits
(4) Initiates sprouting of potato tubers

19. During removal of shoot tips in tea plant, the most active hormone is
(1) Cytokinins
(2) Auxins
(3) Ethylene
(4) Gibberellins

20. Choose the mismatched pair.
(1) Ethephon – Thinning of cherry
(2) Auxins – Xylem differentiation
(3) ABA – Mostly antagonist to GA_3
(4) Cytokinins – Bolting

21. Growth in plant organs is
(1) Qualitative and extrinsic
(2) Quantitative and intrinsic
(3) Qualitative and intrinsic
(4) Quantitative and extrinsic

22. The growth in plants differs from growth in animals in
(1) Being localized and indefinite
(2) Being indefinite
(3) Having indefinite lifespan
(4) Having definite lifespan

23. Find out the correct statement(s).
(a) Growth in plants is internal/intrinsic and open ended
(b) Formation of cellular materials is called real or protoplasmic growth
(c) Plant growth is diffused only during the early embryonic stage
(1) Only (a) & (b)
(2) (b) only
(3) Only (b) & (c)
(4) (a), (b) & (c)

24. Find odd one w.r.t. differentiation
(1) Loss of nucleus in sieve
(2) Death of protoplasm in tracheary elements
(3) Callus formation
(4) Lignification in vessels

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25. Which of the given is/are examples of differentiation?
(1) Loss of nucleus and perforations in some sieve tube members
(2) Lignin deposition in tracheids and vessels
(3) Differential wall thickening in guard cells
(4) More than one option is correct

26. Larkspur, *Ranunculus* and cotton plants are similar in the presence of
(1) Developmental heterophylly
(2) Plasticity
(3) Environmental heterophylly
(4) Homospory

27. If an etiolated stem could be first saturated with auxin by spraying and then exposed to a streak of light from one side, it will
(1) Bend towards the light
(2) Bend away from the light
(3) Grow straight upwards
(4) Be prevented from growing

28. Which is not a physiological effect of auxins?
(1) Cell elongation
(2) Development of parthenocarpic fruits
(3) Prevention of abscission of leaves and fruits
(4) Reversal of genetic dwarfism

29. The direction of the transport of auxins is
(1) Polar in basipetal (2) Polar in acropetal
(3) Through xylem (4) Through phloem

30. Pruning of tea plants is done to discard the effects of
(1) GA_3
(2) Auxin
(3) Cytokinin
(4) ABA

31. Which of the following hormone is weak organic acid having unsaturated ring structures and derived from amino acid?
(1) Cytokinin (2) Auxin
(3) Gibberellins (4) Ethylene

32. Specific property attributed to GA is
(1) Shortening of genetically tall plants
(2) Elongation of genetically dwarf plants
(3) Rooting of stem cuttings
(4) Promotion of leaf and fruit fall

33. The plant hormone which is basic in nature?
(1) Auxin (2) Gibberellins
(3) Cytokinin (4) Abscisic acid

34. Delay of senescence or Richmond Lang effect is a physiological effect of
(1) IAA (2) CK
(3) GA (4) C_2H_4

35. Shelf life of vegetables and cut flowers can be increased by commercial application of
(1) Cytokinin (2) AMO1618
(3) Cyclocel (4) Phosphon-D

36. The phytohormone combination which is the key regulator of cell differentiation and morphogenesis is
(1) Cytokinin & IAA
(2) IAA & ABA
(3) IAA & GA_3
(4) Cytokinin & Gibberellin

37. Cytokinins are said to be antiageing hormone because they delay the senescence by
(1) Controlling mobilisation of resources
(2) Controlling protein synthesis
(3) Decreased morphogenesis and high respiration
(4) Both (1) & (2)

38. Number of female flowers can be increased by application of
(1) IAA
(2) C_2H_4
(3) CK
(4) All of these

39. Triple response is shown by hormone
(1) Ethylene
(2) CK
(3) 2, 4-D
(4) GA_3

40. $CH_2 = CH_2$ is mainly responsible for
(1) Formation of internode
(2) Formation of nodes
(3) Ripening of fruits
(4) Formation of internodes

41. The hormone which can replace long days and low temperature requirement for flowering in some plants is
(1) Gibberellin
(2) Cytokinin
(3) Vernalin
(4) Ethylene

42. Gibberellin mediate amylase formation during germination of cereal grains is inhibited by
(1) Abscisic acid
(2) Ethylene
(3) Gibberellins
(4) Cytokinins

43. Select an incorrect match
(1) Tryptophan – Auxin
(2) Methionine – Ethylene
(3) tRNA – Cytokinin
(4) Violaxanthin – GA_3

44. Which is not true for abscisic acid?
(1) Acts as antitranspirant
(2) Synthesized in chloroplast from carotenoids
(3) Increases stress tolerance in plants
(4) Induces epinasty of leaves and flowers

45. Which hormone stimulates the closure of stomata in the epidermis and increases the tolerance of plants to various kinds of stresses?
(1) ABA (2) Cytokinin
(3) GA_3 (4) Auxin

46. Match the following

Column I		Column II
a. Auxin	(i)	Root hair formation
b. Cytokinin	(ii)	Seed development
c. Ethylene	(iii)	Xylem differentiation
d. ABA	(iv)	Nutrient mobilisation

(1) a(iv), b(ii), c(iii), d(i)
(2) a(ii), b(iii), c(i), d(iv)
(3) a(i), b(iii), c(ii), d(iv)
(4) a(iii), b(iv), c(i), d(ii)

47. Condition of suspended growth due to external environmental conditions is called
(1) Dormancy
(2) Rest
(3) Quiescence
(4) All of these

48. Seed dormancy in tomato seeds is due to
(1) Impermeable seed coat
(2) Immature embryo
(3) Presence of ferulic acid in pulp
(4) Abscisic acid in pulp

49. Which of the given event does not happen during seed germination?
(1) Emergence of radicle
(2) Increase in rate of respiration
(3) Hydrolysis of stored polysaccharides and proteins
(4) Photosynthesis by cotyledons

50. For flowering, critical dark period should always be exceeded in
(1) Long day plants
(2) Short day plants
(3) Day neutral plants
(4) All type of plants

51. Which of the following statements does not characterize photoperiodism?
(1) Mediated by florigen hormone
(2) Conversion of shoot apex into reproductive apex
(3) Red light is stimulatory in SDP, LDP and DNP for flowering
(4) Photoperiodic stimulus is perceived by mature leaf

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52. Hypothetical 'florigen' could be produced prematurely in long day plant by exposing it to
(1) Shortening light period
(2) Red light during night
(3) Extending dark period
(4) Far red light

53. All given statements are correct w.r.t. photoperiodism, except
(1) Discovered by Garner and Allard in a variety of tobacco
(2) Stimulus is received by young leaves
(3) Florigen is the hypothetical hormone produced in response to stimulus
(4) Florigen is transported to bud through phloem

54. Which of the following pair is mismatched?
(1) *Triticum aestivum* – LDP
(2) *Zea mays* – DNP
(3) *Glycine max* – SDP
(4) *Raphanus sativus* – SDP

55. Which is not a requirement of vernalisation?
(1) Aerobic condition (2) Moisture
(3) Low temperature (4) Differentiated tissues

❑ ❑ ❑

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Chapter-12 : Plant Growth and Development

EXERCISE

1.	(4)	2.	(2)	3.	(2)	4.	(2)	5.	(2)	6.	(1)	7.	(1)
8.	(4)	9.	(1)	10.	(2)	11.	(4)	12.	(4)	13.	(4)	14.	(2)
15.	(3)	16.	(2)	17.	(2)	18.	(2)						

ASSIGNMENT

Section - A : NCERT Based MCQs

1.	(4)	2.	(3)	3.	(1)	4.	(4)	5.	(4)	6.	(1)	7.	(2)
8.	(4)	9.	(2)	10.	(3)	11.	(3)	12.	(1)	13.	(3)	14.	(3)
15.	(3)	16.	(4)	17.	(2)	18.	(4)	19.	(3)	20.	(3)	21.	(2)
22.	(4)	23.	(3)	24.	(2)	25.	(3)	26.	(2)	27.	(3)	28.	(3)
29.	(1)	30.	(2)	31.	(1)	32.	(3)	33.	(1)	34.	(4)	35.	(1)
36.	(3)	37.	(1)	38.	(4)	39.	(2)	40.	(4)				

Section - B : Objective Type Questions

1.	(1)	2.	(2)	3.	(2)	4.	(3)	5.	(1)	6.	(3)	7.	(1)
8.	(1)	9.	(1)	10.	(1)	11.	(2)	12.	(1)	13.	(1)	14.	(3)
15.	(3)	16.	(3)	17.	(3)	18.	(1)	19.	(2)	20.	(4)	21.	(2)
22.	(4)	23.	(3)	24.	(3)	25.	(4)	26.	(3)	27.	(1)	28.	(1)
29.	(2)	30.	(4)										

Section - C : Previous Years Questions

1.	(4)	2.	(3)	3.	(3)	4.	(2)	5.	(1)	6.	(4)	7.	(4)
8.	(1)	9.	(4)	10.	(4)	11.	(1)	12.	(4)	13.	(4)	14.	(4)
15.	(2)	16.	(4)	17.	(4)	18.	(1)	19.	(3)	20.	(2)	21.	(4)
22.	(4)	23.	(2)	24.	(2)	25.	(1)	26.	(3)	27.	(1)	28.	(2)

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29. (3) 30. (2) 31. (3) 32. (3) 33. (2) 34. (1) 35. (4)
36. (3) 37. (4) 38. (4) 39. (3) 40. (3) 41. (3) 42. (2)
43. (1) 44. (3) 45. (1) 46. (2) 47. (4) 48. (1) 49. (3)
50. (4) 51. (2) 52. (4) 53. (4) 54. (1) 55. (2) 56. (4)
57. (4) 58. (1) 59. (3) 60. (2) 61. (2) 62. (4) 63. (1)
64. (3) 65. (1) 66. (1) 67. (2) 68. (4) 69. (1) 70. (4)
71. (4) 72. (1) 73. (1) 74. (4) 75. (1)

Section - D : NEET Booster Questions

1. (2) 2. (3) 3. (1) 4. (1) 5. (4) 6. (2) 7. (1)
8. (4) 9. (2) 10. (1) 11. (4) 12. (4) 13. (2) 14. (1)
15. (3) 16. (4) 17. (3) 18. (3) 19. (1) 20. (4) 21. (2)
22. (1) 23. (4) 24. (3) 25. (4) 26. (2) 27. (1) 28. (4)
29. (1) 30. (2) 31. (2) 32. (2) 33. (3) 34. (2) 35. (1)
36. (1) 37. (4) 38. (4) 39. (1) 40. (3) 41. (1) 42. (1)
43. (4) 44. (4) 45. (1) 46. (4) 47. (3) 48. (3) 49. (4)
50. (2) 51. (3) 52. (2) 53. (2) 54. (4) 55. (4)

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